BioCoLoop: Collaborative Agentic Research for Biological Model Improvement

Anonymous authors
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Abstract

Biological model development requires both fitting parameters and deciding which model configurations to investigate. When measurements are held by different laboratories, these decisions must draw on distributed training and evaluation. We introduce BioCoLoop, a collaborative research framework that connects laboratory-local fitting to a shared record of candidate designs, development trajectories and selection decisions. Task adapters implement this interface for drug–target interaction prediction, observed-response proteomic efficacy prediction and cell perturbation identification. We evaluate three distinct uses of collaboration: increasing laboratory participation, conditioning proposals on prior development outcomes, and allocating training from early learning trajectories. Source-defined laboratory experiments measure transfer across biological studies, while same-access search controls separate proposal quality from access to additional data. The combined framework improves four of five primary endpoints relative to the single-laboratory fixed model; the effect of proposal history is task-dependent. A separate allocation study improves double-gene perturbation identification by 10.84 percentage points over uniform allocation at matched training and evaluation counts, although simpler promotion also recovers much of this gain. These results establish a common experimental framework for collaborative biological model development and identify training allocation and development-to-test generalization as consequential research decisions.

1. Introduction

Artificial intelligence supports biological research through predictive modeling and the generation and evaluation of hypotheses (Cui et al., 2024; Theodoris et al., 2023; Roohani et al., 2024). Self-improving agents automate parts of model development (Jiang et al., 2025b; Lu et al., 2024; Zhou et al., 2026) through an iterative process of proposing, testing and revising (Madaan et al., 2023; Shinn et al., 2023; Huang et al., 2023). An agent proposes changes to a model design or its training hyperparameters, trains and evaluates the candidate, and uses the results to inform the next proposal.

In biology, useful data are often distributed across laboratories studying different cell types, perturbations or experimental assays. Data-sharing constraints can prevent these measurements from being pooled (Sheller et al., 2020; Bakhtiari et al., 2025; Dayan et al., 2021; Kaissis et al., 2020; Rieke et al., 2020). Collaborative learning enables laboratories to train a shared model through locally computed updates (McMahan et al., 2017; Wang et al., 2024; Li et al., 2020; Karimireddy et al., 2020; Kairouz et al., 2021). Automated federated design already combines configuration or system changes with distributed fitting (Khodak et al., 2021; Li et al., 2026). The biological question is how to connect these decisions to measurements from different assays and experimental contexts, and how to distinguish the value of additional data from the value of the research procedure.

Local development evidence provides a complementary research signal: it records how candidate configurations perform, how their training trajectories evolve and which trials succeed or fail. Sharing these summaries allows laboratories to inform subsequent experiments while retaining their measurements locally. *Can an iterative research process use laboratory-local training and evaluation to guide model development without pooling the underlying data?*

BioCoLoop extends collaboration from parameter fitting to model development by sharing both model updates and laboratory-local development evidence. Figure 1 places this framework along-

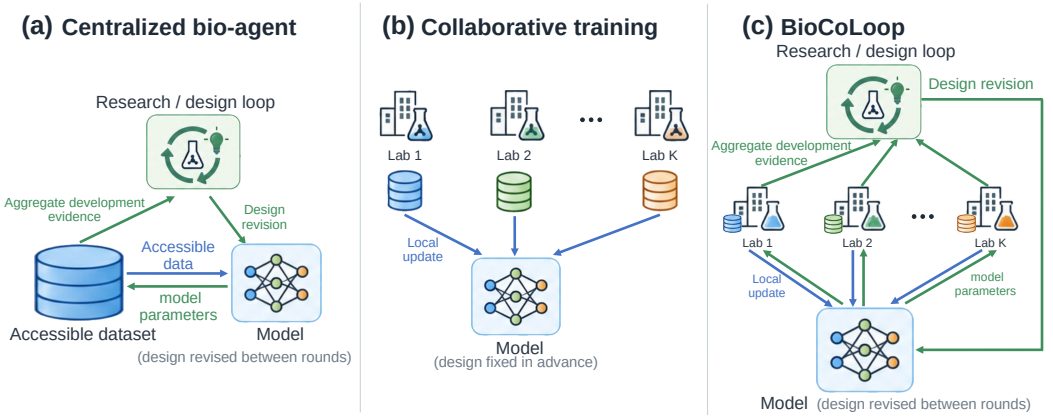


Figure 1: Research settings for biological AI. A centralized bio-agent revises a model using centrally accessible data. Collaborative training allows K laboratories to optimize the same model design and training hyperparameters by sharing model updates rather than raw data. BioCoLoop additionally uses results from local evaluation to guide subsequent changes to the model design and training hyperparameters.

side centralized self-improving agents and conventional collaborative training. An inner loop fits and evaluates candidates across laboratories. An outer loop uses the resulting evidence to revise proposals. Collaborative participation supplies training data and evaluation panels; development-history feedback uses their measured outcomes to guide search. We also evaluate a separate policy that uses early development trajectories to allocate further training.

We evaluate BioCoLoop on three biological task families with two proposal models, conventional search controls and task-adapted research-agent baselines. The evaluation separates an access scenario—ten participating laboratories versus one—from same-access tests of search and allocation. Our contributions are:

- A common biological research interface connects local fitting, development evaluation and executable configuration proposals across interaction, proteomic and perturbation tasks.
- A shared experiment record supports two separately evaluated decisions: choosing the next configuration and allocating further training from early scores and improvement trajectories.
- Controlled experiments quantify laboratory-participation gains, cross-study transfer and matched-budget training allocation, and test how proposal feedback translates development evidence into held-out model performance.

2. Related work

Collaborative learning for biological data. Parameter aggregation enables institutions to train a shared model while retaining their measurements locally (McMahan et al., 2017; Sheller et al., 2020). In single-cell analysis, scFed addresses cell-type classification and FedscGen addresses distributed batch correction (Wang et al., 2024; Bakhtiari et al., 2025). Tabula and its Chiron implementation support collaborative single-cell foundation-model training through an agent-enabled platform (Ding et al., 2026; Mechtel et al., 2026). Automated design can interact with local training: FedEx tunes hyperparameters while training shared weights (Khodak et al., 2021), and Helmsman synthesizes collaborative-learning systems through planning, coding and simulation-based refinement (Li et al., 2026).

Self-improving agents in biological research. Self-improving agents turn experiment outcomes into subsequent computational proposals. AIDE searches candidate programs through drafting, debugging and improvement (Jiang et al., 2025b), while the AI Scientist and AI-Scientist-v2 connect ideas to executed experiments, replanning, tree search and writing (Lu et al., 2024; Yamada et al., 2025). DrugEvolve combines research, implementation, evaluation and algorithm memory for drug-

development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). BioCoLoop studies how multiple data-owning laboratories can supply development feedback through a shared research interface. TAPB, a ProteinTalks-derived efficacy head and scDEBART-based cell representations provide its task-specific prediction components (Lin et al., 2025; Sun et al., 2026; Sung & Kim, 2026). The proposal schema, fitting service and selection mechanism are common across tasks. Strong simple references on some perturbation benchmarks motivate evaluating the research procedure with fixed task predictors (Ahlmann-Eltze et al., 2025). Configuration search also connects to random search (Bergstra & Bengio, 2012), Bayesian optimization (Snoek et al., 2012) and language-model proposal distributions (Yang et al., 2024). We use feedback-free direct search to isolate the effect of development history under a shared proposal and fitting interface.

Collaborative scientific agents. The Virtual Biotech coordinates specialist agents around therapeutic-discovery evidence (Zhang et al., 2026). AutoScientists organizes experimentation through shared state and records successful and failed directions (Gao et al., 2026). Helmsman directly addresses agent-generated federated systems, including simulation-based diagnosis and revision (Li et al., 2026). BioCoLoop's focus is a common biological predictor-development interface: laboratories supply local fits and development measurements, and controlled experiments separate participation, proposal history and training allocation. Its finite configuration library supports these comparisons; it differs from open-ended federated-system code synthesis.

3. Method: collaborative biological model improvement

3.1 Problem setting and notation

Consider K laboratories working on the same biological prediction task. Laboratory i holds training data D_i^{train}, with $n_i = |D_i^{\text{train}}|$. Evaluation uses laboratory-local development panels $\{D_i^{\text{dev}}\}_{i\in\mathcal{P}}$, where $\mathcal{P}$ indexes the panels specified by the evaluation protocol. A task specification $\mathcal{T}$ defines the prediction inputs, outputs, training loss and evaluation metric. We write the task predictor as $f_{a,w}$, where $a \in \mathcal{A}$ specifies its design and w its trainable parameters. Given an initial design, a design space $\mathcal{A}$ and a research budget, BioCoLoop seeks a design and fitted predictor that perform well on the designated evaluation population. The three task families instantiate $f_{a,w}$ as task-specific predictors within this shared research procedure.

In the main experiments, the *registered executable design space* $\mathcal{A}$ contains complete candidate configurations with four editable choices: learning rate, weight decay, coordinator momentum and an optional residual prediction module. Each `design_id` selects one configuration. The architecture choice enables or disables a width-32 residual module with a zero-initialized output layer; the task adapter supplies the backbone and prediction head. Appendix E.6 defines a separate, larger-data mechanism study with its own design space and training schedule.

A *shared coordinator* is the trusted execution service used by all participating laboratories. It resolves each design identifier, initializes and broadcasts the predictor, coordinates local fitting, aggregates model states and bounded development measurements, and applies the checkpoint and design-selection rules. Laboratories receive the same candidate state and schedule while retaining raw training and development measurements locally. The coordinator supplies the aggregate evidence to research model S (Figure 2), a language model whose weights θ remain fixed throughout a run. We use this fitting interface to evaluate history-guided proposal revision (Section 3.3) and trajectory-guided training allocation (Section 3.4) separately.

3.2 Collaborative fitting and shared evaluation

For each candidate a, the coordinator resets the predictor to its prescribed initialization w_0 and distributes it to all training laboratories. Let LocalEpoch_a denote one full local AdamW epoch with the candidate's learning rate and weight decay, batch size 64 and gradient-norm clipping at one. The optimizer state resets at each round. With sample weights $\pi_i = n_i / \sum_{j=1}^{K} n_j$, coordinator momentum

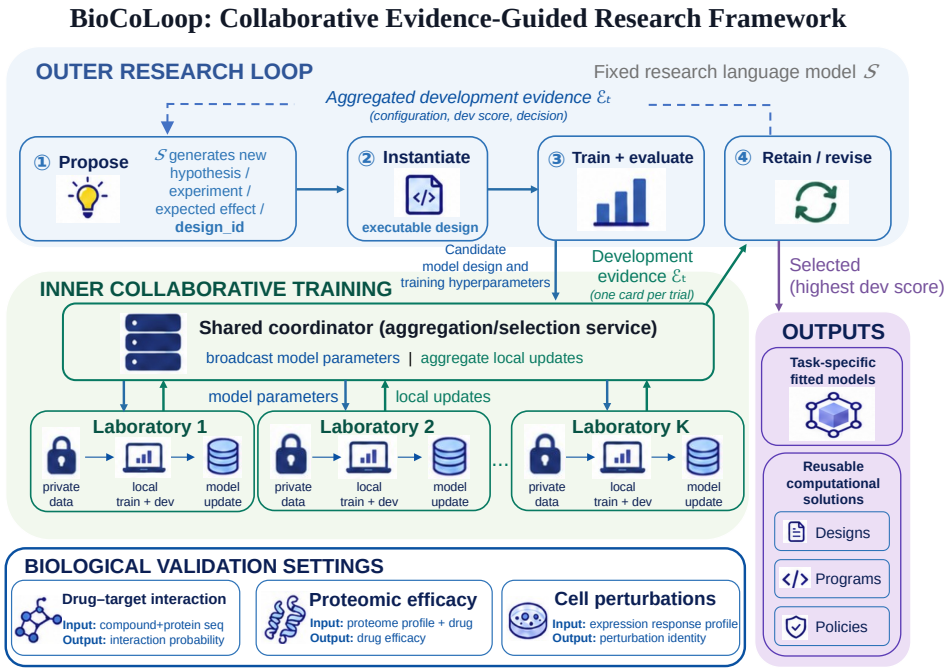


Figure 2: BioCoLoop's common research interface. Research model S proposes an executable configuration; the coordinator fits it using updates from K laboratories and summarizes development evaluation in an evidence card. The ordered history $\mathcal{E}_t$ conditions the next proposal. Task adapters supply biological inputs, prediction heads and losses. Sections 3.2–3.3 define fitting, evidence and selection; Section 3.4 evaluates training allocation separately.

coefficient β_a and initial velocity $v_0 = 0$, round r updates the trainable parameters by

$$u_{i,r} = \mathrm{LocalEpoch}_a(w_{r-1}; D_i^{\mathrm{train}}), \quad \bar{w}_r = \sum_{i=1}^{K} \pi_i u_{i,r}, \\ v_r = \beta_a v_{r-1} + \bar{w}_r - w_{r-1}, \qquad w_r = w_{r-1} + v_r. \tag{1}$$

The coordinator carries momentum across aggregation rounds through v_r. With $\beta_a = 0$, aggregation reduces to sample-weighted parameter averaging (McMahan et al., 2017). In each round, the coordinator sends the current trainable state and fixed candidate settings to every laboratory. Laboratory i returns its locally fitted state $u_{i,r}$ and training-example count n_i. Frozen feature extractors remain unchanged. Task adapters supply the biological inputs, trainable prediction components and local loss; raw examples, expression profiles, labels and per-example identifiers remain local.

Every five rounds and at the final round, laboratories evaluate the aggregated predictor f_{a,w_r} on the development panels indexed by $\mathcal{P}$. Each returns a scalar primary score $s_i(a, r)$, development loss $\ell_i^{\mathrm{dev}}(a, r)$ and the bounded trajectory diagnostics required by the protocol. The main comparison uses participating laboratories' panels; fixed-pool repartitioning retains all ten original panels; source-transfer studies use the original target panels. The coordinator computes

$$\bar{s}(a, r) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} s_i(a, r), \qquad \bar{\ell}(a, r) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \ell_i^{\mathrm{dev}}(a, r). \tag{2}$$

The selected round $\widehat{r}_a$ maximizes $\bar{s}(a, r)$, with lower $\bar{\ell}(a, r)$ breaking ties and the earlier round retained on a complete tie. Its parameters are $\widehat{w}_a = w_{\widehat{r}_a}$. The pair $q(a) = (\bar{s}(a, \widehat{r}_a), -\bar{\ell}(a, \widehat{r}_a))$ ranks the design. We call $\bar{s}(a, \widehat{r}_a)$ the *dev score*, labelled "highest dev score" in Figure 2. Training weights examples, while selection weights the designated development panels equally. The evaluation schedule assigns each candidate 100 training rounds in the main proposal-search experiments. Each trial produces a *development-evidence card*. After t trials, the sequence $\mathcal{E}_t = (e_1, \ldots, e_t)$

forms the accumulated *development evidence*. For $1 \le j \le t$, card e_j records the executed configuration a_j, hypothesis h_j, best development primary score and loss (s_j, ℓ_j), selected round $\widehat{r}_j$ and acceptance decision dec_j. Endpoint diagnostics Δ_j contain the first and last development-evaluation rounds, their sample-weighted training losses, equal-laboratory development scores and losses, and the changes between those measurements, plus the optional worst-laboratory score and evaluating-laboratory count.

For auxiliary-source studies with different prediction targets, shared encoders follow the same update rule, while each source-specific head is aggregated only over laboratories using that prediction target. Checkpoint selection uses the original target's development panels. Each source retains its own prediction objective while contributing updates to the shared representation.

3.3 Proposal revision from development evidence

The first proposal is generated once without development feedback and shared by direct and loop search. History feedback begins with the second proposal. For subsequent research steps t, let a_t be the retained design, $\mathcal{E}_t$ the development evidence accumulated so far and $\mathcal{U}_t$ the untried configurations. The research model S, with weights θ held fixed, generates a proposal p_t and the coordinator maps its `design_id`, denoted $\mathrm{id}(p_t)$, to the candidate:

$$p_t \sim P_\theta(\cdot \mid \mathcal{T}, a_t, \mathcal{E}_t, \mathcal{U}_t), \qquad \widetilde{a}_t = \mathrm{Compile}(\mathrm{id}(p_t)). \tag{3}$$

The proposal also contains a `hypothesis`, an `experiment` and an `expected_effect`, recording the motivation, intended comparison and anticipated outcome. The identifier determines execution; measured development performance determines selection.

After training $\widetilde{a}_t$, the coordinator constructs the next card e_{t+1}. Comparing q lexicographically by higher score and then lower loss gives

$$a_{t+1} = \begin{cases} \widetilde{a}_t, & q(\widetilde{a}_t) >_{\mathrm{lex}} q(a_t), \\ a_t, & \text{otherwise}, \end{cases} \qquad \mathcal{E}_{t+1} = \mathcal{E}_t \parallel [e_{t+1}], \tag{4}$$

where $\parallel$ appends a card to the history. Unsuccessful trials remain in the history. Invalid proposals and numerically failed fits consume a proposal slot and leave the incumbent unchanged. Updated context carries these outcomes into subsequent proposals. After the proposal budget is exhausted, the retained design and its selected checkpoint are evaluated on the held-out task.

3.4 Evidence-guided training allocation

We use trajectory-guided allocation in a controlled study (Section 4.4) to test whether early evaluation evidence improves the use of a fixed training budget. The screening–promotion policy retains both high-scoring and still-improving candidates. It screens ten predetermined designs for 20 rounds, evaluating every five rounds through Equation 2. Within each of three learning-rate groups, it promotes the best-scoring design and, among the remaining designs, the one with the largest score improvement from round 5 to round 20. Development loss breaks score ties. The six promoted designs restart from their initial parameters for 100-round training, and their best development checkpoints determine the final predictor.

The two promotion criteria reserve training capacity for candidates at different stages of convergence. Current score favors a strong screening checkpoint, while improvement over the shared screening interval retains a candidate whose score is still rising. Grouping by learning rate preserves representation from each group in the full-training stage. All promotion decisions use development measurements.

The control trains all ten designs for 80 rounds. Both schedules use 800 aggregation rounds and 160 aggregate development evaluations. This study evaluates allocation over a predetermined candidate set independently of proposal generation. The main comparison and proposal-history experiments use 100 training rounds per candidate. Section 4.4 reports the allocation results; Appendix A provides the complete proposal schema, algorithm and allocation rules.

4. Experiments

4.1 Tasks, datasets and evaluation

We evaluate BioCoLoop on three biological prediction problems with task-specific base models and a common procedure for collaborative training, evaluation and proposal generation. The main setting partitions each dataset into ten disjoint groups. Complementary studies use distinct biological datasets or experimental scenarios as laboratories to evaluate source transfer.

Drug–target interactions. Given a compound and a protein sequence, the model predicts whether they interact. BindingDB provides 50,149 training pairs and 5,604 development pairs, with canonical compounds defining laboratory assignments. We use TAPB as the base model, with frozen ESM2 protein features and trainable drug, fusion and classification modules (Lin et al., 2025). The primary metric is AUROC on 5,505 random test pairs. Additional test sets containing 3,791 unseen-drug pairs and 2,507 unseen-protein pairs measure generalization to unseen entities. AP, MCC, accuracy and cross-entropy are reported as secondary metrics.

Observed-response proteomics. Given a drug representation and its measured 6h and 24h proteomic responses, the model predicts binary efficacy. A ProteinTalks PTPC subset provides 386 training, 105 development and 148 test conditions (Sun et al., 2026). Compound-group assignment gives each laboratory 49-50 training and development conditions. The base model is a ProteinTalks-derived efficacy model with protein and drug branches, trained from random initialization. Average precision (AP) is the primary metric. AUROC, accuracy and cross-entropy are secondary metrics. Unlike de novo response prediction, this task conditions on observed post-treatment proteomic measurements.

Cell perturbation identification. Given an expression response and five candidate interventions, the model identifies the intervention whose predicted response has the highest cosine similarity to the observation. We use VCC single-gene perturbations (Arc Institute, 2025), Norman double-gene perturbations (Norman et al., 2019) and a Tahoe subset of drug perturbations in A549 at 5 μM (Zhang et al., 2025). Intervention-disjoint training, development and test counts are 60/20/20, 70/20/27 and 180/60/67, respectively. The base model uses a frozen scDEBART backbone with a trainable response head (Sung & Kim, 2026). All methods use the same corrected attention masks and the same candidate sets. The primary metric is intervention-macro Top-1, which first averages accuracy within each intervention and then across interventions. Random selection has an expected Top-1 of 20%. Micro Top-1, MRR, Top-3 and cross-entropy are reported as secondary metrics. Appendix A provides the response construction, chemical conditioning, training losses, and secondary evaluation details.

4.2 Compared methods and shared protocol

We compare BioCoLoop with three types of baselines. The *fixed model* fits the task-specific base model once with a predefined configuration. *Direct optimization* uses BioCoLoop's proposal generator and registered executable design space, generating proposals without previous development outcomes. AI-Scientist-v2 (Yamada et al., 2025) and AI-Researcher (Tang et al., 2025) are adapted to the same biological tasks and fitting interface. Every method returns one predictor; laboratories contribute to its fitting, not an inference-time ensemble.

BioCoLoop combines collaborative fitting and evaluation with development-history feedback. We use Qwen2.5-7B-Instruct and GPT-5.6 Luna as proposal generators (OpenAI, 2026). Table 1 is an *access-scenario comparison*: BioCoLoop uses ten laboratories and the other methods use laboratory 0. It compares complete workflows with different available data and local computation. The same-access experiments in Section 4.7 and Appendix E instead fix all ten training laboratories, development panels, design choices and candidate-training budgets to examine search itself. Here *direct* means feedback-free proposal generation, not a single training run; the fixed-model control is the single-fit reference.

The main comparison uses twelve configurations and at most six proposals beyond the initial design. Each valid candidate starts from the prescribed task initialization and receives 100 training rounds. Development score and loss determine the retained design and checkpoint before held-out scoring. We report means and sample standard deviations over seeds 42–44 on fixed data partitions.

Table 1: Access-scenario comparison: ten participating laboratories versus one. Scores $\times 100$ are mean $\pm$ sample SD over seeds 42–44. Data and local computation differ with access; this table measures complete workflows, not the isolated effect of proposal history. Bold and underline indicate the best and second-best complete scores within each block. †: two scored seeds; –: no sealed score; ⋆: extended-context adaptation. Execution outcomes are in Appendix C; same-access search is reported separately.

Method	Labs	DTI	Proteomics	Cell perturbation (Top-1)		
		AUROC ↑	AP ↑	VCC ↑	Norman ↑	Tahoe ↑
Qwen2.5-7B-Instruct						
Fixed model	1	82.83 ± 1.22	<u>25.05</u> ± 4.56	<u>29.94</u> ± 0.00	**25.56** ± 0.00	<u>19.40</u> ± 0.00
Direct	1	<u>89.89</u> ± 1.02	20.13 ± 0.07	21.09 ± 4.38	14.15 ± 0.00	15.42 ± 6.21
AI-Scientist-v2	1	89.68 ± 0.80	20.18 ± 0.43	18.61 ± 2.78	14.12 ± 0.04	15.42 ± 6.21
AI-Researcher	1	86.09 ± 0.88	20.12 ± 0.05	19.70 ± 2.87†	21.73 ± 6.63	17.41 ± 6.21
BioCoLoop	10	**93.76** ± 1.11	**36.99** ± 0.22	**31.54** ± 0.08	<u>22.25</u> ± 0.45	**22.39** ± 2.59
GPT-5.6 Luna (low reasoning)						
Fixed model	1	82.83 ± 1.22	<u>25.05</u> ± 4.56	<u>29.94</u> ± 0.00	**25.56** ± 0.00	<u>19.40</u> ± 0.00
Direct	1	<u>89.89</u> ± 1.02	20.17 ± 0.42	18.61 ± 2.78	21.75 ± 6.59	15.92 ± 6.03
AI-Scientist-v2	1	89.59 ± 0.95	23.37 ± 5.89	18.61 ± 2.78	14.12 ± 0.04	11.94 ± 2.11†
AI-Researcher	1	–	20.29 ± 0.27⋆	18.61 ± 2.78⋆	14.12 ± 0.04⋆	15.42 ± 6.21⋆
BioCoLoop	10	**93.67** ± 0.98	**37.11** ± 0.01	**31.50** ± 0.00	<u>22.69</u> ± 0.73	**22.39** ± 2.59

These summaries describe training and search variability; paired cluster-bootstrap intervals in the Statistical Supplement quantify held-out sample uncertainty. The allocation, source-transfer and extended-search studies use their separately specified budgets and seeds.

4.3 Performance with broader laboratory participation

Table 1 compares complete workflows under the specified access scenario. BioCoLoop improves four of five primary endpoints over the single-laboratory fixed model in both proposal-model blocks. The fixed model retains the highest Norman Top-1. The explicit laboratory-count column distinguishes these access gains from the same-access search effects in Section 4.7.

With Qwen, BioCoLoop reaches 93.76 AUROC on BindingDB and 36.99 AP on proteomics, compared with fixed-model scores of 82.83 and 25.05; Luna reaches 93.67 and 37.11. BioCoLoop also leads the complete VCC and Tahoe comparisons in both blocks. Norman is the exception: the fixed model scores 25.56, versus 22.25/22.69 for Qwen/Luna BioCoLoop.

Single-laboratory random search, Bayesian optimization and twelve-design enumeration also remain below the ten-laboratory Qwen workflow (Appendix D). They test whether extra search compensates for limited data access; the ten-laboratory controls in Appendix E address search quality. Appendix E.7 additionally measures fixed-model variation across all ten single-laboratory identities.

4.4 Evidence-guided training allocation

Evidence-guided allocation improves held-out performance at a matched budget (Table 2). Uniform allocation trains all ten designs for 80 rounds. The evidence-guided policy screens them for 20 rounds and restarts six selected designs for 100-round training. Each schedule uses 800 aggregation rounds and 160 development evaluations.

Norman Top-1 increases from 11.48% to 22.32%, a 10.84-point gain with a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. A selected low-learning-rate model improves beyond the uniform stopping point. Across the twelve task–seed pairs, allocation yields four wins, eight ties and no losses: three gains on Norman and one on Tahoe. Screening and restarting costs are included. This fixed-slate experiment isolates training allocation from language-model proposal history.

A fresh 360-fit study adds score-only and random promotion (Table E.5). On Tahoe, trajectory promotion reaches 22.89 versus 20.90 for score-only and 21.95 for random. On Norman, score-only reaches 21.70 and random 22.63, versus our 22.32; random also leads on VCC (22.49 versus 20.01).

Table 2: Evidence-guided training allocation at a matched budget. Held-out scores use the same ten designs, ten laboratories, 800 full-client aggregation rounds and 160 aggregate development evaluations per method. Uniform allocation gives every design 80 rounds; the evidence-guided scheduler screens all designs for 20 rounds and restarts six promoted designs for 100-round training. Gain is evidence-guided minus uniform, in percentage points; W/T/L counts seed-level wins, ties and losses (total 4/8/0). Norman 95% CI: [4.54, 17.73]; Tahoe: [−1.00, 7.46]. The separate DTI development replay is in Appendix A.

Benchmark	**Uniform**	**Evidence-guided**	**Gain**	**W/T/L**
Proteomics: efficacy	**37.47**	**37.47**	+0.00	0/3/0
Cell: VCC single-gene	**20.01**	**20.01**	+0.00	0/3/0
Cell: Norman double-gene	11.48	**22.32**	**+10.84**	3/0/0
Cell: Tahoe drug	20.90	**22.89**	**+1.99**	1/2/0

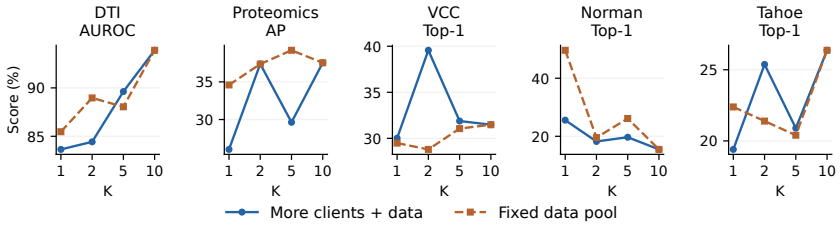


Figure 3: Effect of the number of laboratories. Solid lines add both laboratories and training data. Dashed lines divide the same training data across different numbers of laboratories. Full results are reported in Appendix B.

Selective longer fitting therefore explains much of the gain over uniform allocation, while the choice of promotion rule has task-dependent effects.

4.5 Learning from different data sources

Source-transfer studies use laboratories from distinct biological datasets or experimental settings. We combine BindingDB with BioSNAP, Davis and Human; VCC with three external single-cell studies; and PTPC with Lin, Ruprecht and decryptE (Lin et al., 2026; Ruprecht et al., 2020; Eckert et al., 2025). Target development data select each model, and all configurations use the same target test set.

With one proteomic study per laboratory, the three auxiliary sources raise target AP from 34.41 to 37.53 and AUROC from 72.33 to 74.39. Shared encoders combine local updates while source-specific heads retain their own objectives. Davis and Human also improve target DTI AUROC. Transfer depends on source compatibility: at four laboratories and 60 training interventions in both cell arms, same-source VCC reaches 30.70% Top-1 versus 28.82% with external studies. Appendix B reports every source comparison; Appendix E.4 tests proposal history at matched source access.

4.6 Effect of the number of laboratories

We vary $K \in \{1, 2, 5, 10\}$ by either adding laboratories and training data or repartitioning a fixed total data pool. Adding data through participation can improve performance, while fixed-pool repartitioning has task-dependent effects (Figure 3). Proteomics performs best at an intermediate K, and Norman degrades as the fixed pool is split across more laboratories. Performance depends on both data access and the partition used for local training.

Both settings use the default design and training schedule. Increasing participation adds local development panels; fixed-pool repartitioning keeps all ten original evaluation panels. In cell tasks, training groups also define contrastive-negative pools. Appendix B.1 reports these optimization and evaluation differences alongside the complete results.

Table 3: Same-access search: all methods use ten laboratories and the same development panels. Searches use twelve designs, six proposal slots and at most 700 aggregation rounds, including initialization; the fixed reference uses 100 rounds. Scores $\times 100$ are conditional on one training seed per task (DTI 42; others 61), not the three-seed access comparison. Random is its exact policy expectation; variability and actual-fit sensitivities are in Appendix E. Bold marks the best unrounded score, including ties.

Method	DTI AUROC ↑	Proteomics AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
Fixed model	94.27	**37.27**	**31.50**	15.48	25.37
Random, exact expectation	94.45	**37.27**	27.53	22.66	26.59
GP–EI	**94.60**	**37.27**	**31.50**	22.30	25.37
Qwen direct	94.27	**37.27**	**31.50**	22.30	25.37
Qwen history	94.27	**37.27**	**31.50**	22.30	25.37
Luna direct	**94.60**	**37.27**	**31.50**	22.30	**26.87**
Luna history	94.27	**37.27**	**31.50**	**23.11**	**26.87**

4.7 Same-access configuration search

Table 3 fixes ten-laboratory access and adds random search without replacement and Gaussian-process expected improvement (GP–EI) to the paired direct/history study. The retrospective comparison reuses all twelve complete fits per task. Development-only selection is sealed before scoring; random expectations integrate all 462 six-design subsets. Appendix E gives search-policy variability, shared-first-proposal controls and sensitivities matching realized fit counts when proposals fail.

On Norman, Luna history reaches 23.11 Top-1 versus 22.30 for direct/GP–EI and a random expectation of 22.66. DTI instead favors direct Luna and GP–EI; proteomics ties, and VCC's language-model and GP–EI searches retain the fixed reference. Table B.9 reports every short and extended comparison. Its paired trajectory diagnostics describe retrospective target crossings, not realized compute savings.

A separate three-seed study uses 414 development perturbation genes across four source-defined contexts (Appendix E.6). Direct, structured history, random and GP–EI share 16 designs, six requests and 16,940 optimizer steps per search. Structured history reaches 22.89% held-out Top-1 versus 22.62% for direct and 22.67% for random; GP–EI reaches 23.15%. Structured feedback changes the selected design in two seeds, with a positive mean difference from direct whose confidence interval spans zero. Together, the studies establish task-dependent history effects rather than a consistent search advantage.

4.8 Validation under extended search

Validation coverage affects the value of further search. Re-evaluation reproduces the original VCC/Tahoe failures: Qwen/VCC gains 1.17 development points but loses 11.89 test points, while Tahoe gains 5.00 and loses 11.94 (Table E.2). Equal-laboratory and pooled-intervention scores coincide; twenty paired option-roster redraws instead expose selection sensitivity (Table E.3). In the larger-data study, all final searches improve on their fitted references, although local reversals remain (Figure 6). Appendix E separates these findings from untested explanations involving new biological partitions.

5. Conclusion

BioCoLoop extends collaborative biological learning from fitting model parameters to evaluating and revising executable model designs. The shared interface supports three task families, beneficial transfer from compatible external studies, and improved double-gene perturbation identification through selective longer fitting. Controlled comparisons separate these gains from history-specific effects: simple promotion recovers much of the allocation benefit, while proposal history helps in particular settings rather than consistently outperforming conventional search. Development-panel diagnostics identify evaluation coverage as a concrete priority for improving the loop. Together, these results provide a practical framework and an experimental basis for collaborative biological model development.

Reproducibility statement

Section 3 specifies the research and training interfaces; Section 4 defines task inputs, partitions, reference models and metrics. Appendix A gives the proposal schema, design library, optimization settings and allocation rules. Appendices B and C document source/sensitivity studies and public-agent adaptations; D and E separate single-laboratory search from same-access search, allocation and validation studies. The Statistical Supplement reports complete original per-seed contrasts, including non-positive differences. The research records retain code and data hashes, seeds, development trajectories, selection decisions and saved predictions. The original proposal log covers all 504 attempted slots (20×6 short-protocol and 16×24 extended-protocol slots). The revision adds retrospective cache-based search controls, checkpoint diagnostics, and a separate 360-fit allocation comparison, with their own selection seals, source bindings and independent score checks. Random-search variability is distinguished from training-seed variability. Qwen generation is seed-reproducible; Luna and public-agent interface constraints are specified in Appendices A and C.

The fresh history-representation study in Appendix E.6 adds 48 newly fitted candidates and three paired Qwen search repetitions. Its released summaries include all methods, exact random-policy expectations, step/example accounting and every development-selected prefix. The extension archive records source hashes and protocol versions; the earlier parser diagnostic is retained separately from the final scientific comparison.

Ethics statement

This computational study uses previously released biological resources under their source access conditions and collects no new measurements or participant data. Raw data remain local in the simulated laboratory protocol. Multi-institutional deployment would require a privacy and security policy for parameter updates and aggregate diagnostics. Predictions are evaluated for research use and have no biological or therapeutic validation.

AI use statement

The main comparison retains matched proposal records for local Qwen2.5-7B-Instruct and the service identified as `gpt-5.6-luna`. Luna is run at low reasoning effort through a text-only bridge with tools disabled. Its service does not expose a generation seed, so the seed labels identify the training/search repetitions rather than deterministic language-model sampling. AI assistants contributed to hypothesis refinement, experimental protocol and software development, data processing and checks, result analysis and interpretation, literature retrieval, figures, and manuscript drafting and revision. The research artifacts retain the generated proposals and their measured outcomes. The authors are responsible for verifying the final methods, results, citations and manuscript, including AI-assisted material.

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Appendix A. Algorithm and task protocols

How to read the supplementary experiments. Appendix A specifies the algorithm and scorers; B examines laboratory count, source transfer and proposal history; C documents task-adapted public agents; D compares single-laboratory search; and E compares same-access search and training promotion and diagnoses development–test divergence. Ten-versus-one laboratory comparisons measure the complete access scenario, whereas same-access controls isolate research decisions.

Study	What changes	What the comparison measures
Access scenario	One versus ten laboratories and research workflow	Combined data-access and workflow effect
Same-access search	Proposal policy; common data and menu	Configuration-search quality
Source transfer	Biological study supplying local updates	Positive or negative transfer to the target
Training allocation	Which candidates receive longer fitting	Use of a fixed training/evaluation budget
Roster diagnostic	Development distractors and option order	Sensitivity of already selected models

A.1 Research algorithm and optimization

All task families share Section 3's parameter update, development selection and proposal interface. Adapters define task-specific representations, prediction components, losses and scorers. The fitted reference design supplies the first evidence card.

	BioCoLoop research procedure
1	Fit the reference from the task initialization and seed; retain its best development checkpoint and initialize the history.
2	Generate the shared first proposal without development feedback. Subsequent proposals receive the task, untried configurations, retained design and evidence cards.
3	Map `design_id` to its executable configuration, allowing at most two generation attempts per slot.
4	Fit each valid candidate from the same initialization procedure and seed for its assigned aggregation rounds.
5	Select its best development checkpoint; replace the incumbent only for a better score/loss pair.
6	Append the executed configuration, hypothesis, diagnostics and decision, including rejected or failed trials.
7	Return the final retained design, checkpoint and history; evaluate that checkpoint on the held-out task.

Direct and history-guided search share the first feedback-free proposal. Thereafter, direct receives the default configuration and untried identifiers without outcomes; history-guided search receives the retained configuration and accumulated cards. Both share the registered design space, proposal count and training schedule; Appendix B reports every prespecified prefix. A completed configuration fit may be reused only with identical task, data access and seed, retaining its logical position and budget charge.

Each aggregation round runs one local AdamW epoch per participating laboratory, with batch size 64, gradient-norm clipping at one and optimizer-state reset. Designs specify learning rate and weight decay; the proximal penalty is always zero. Parameter aggregation is training-example-weighted, with coordinator momentum applied only to trainable parameters (Equation 1). Floating-point buffers use the same weights; integer and Boolean buffers use elementwise maxima. Candidates share the task-specific public or random initialization procedure and seed. The optional width-32 residual module has zero-initialized output; the original head remains trainable. Code uses *client*/*worker* for laboratory and `server_momentum` for coordinator momentum.

Full fits use 100 rounds and development evaluation every five rounds. Selection maximizes mean primary score, then minimizes mean loss over the designated panels (Section 3.2): the participating laboratories in the main experiment. Complete ties retain the earlier checkpoint or incumbent. Training counts refer to adapter rows—intervention-level profiles, not underlying cell counts, for cell tasks.

A.2 Registered executable design space, proposals and evidence

Each `D` identifier specifies a four-factor configuration below. Starting from D00, the main study selects six distinct designs from D01–D11. Appendix B defines the extended design space.

Design	Learning rate	Weight decay	Momentum	Residual module
D00	0.001	0.001	0	no
D01	0.0001	0.001	0	no
D02	0.01	0.001	0	no
D03	0.001	0	0	no
D04	0.001	0.01	0	no
D05	0.001	0.001	0.9	no
D06	0.001	0.001	0	yes
D07	0.001	0.001	0.9	yes
D08	0.0001	0.001	0	yes
D09	0.0001	0.001	0.9	yes
D10	0.01	0.01	0	no
D11	0.001	0.01	0	yes

Qwen2.5-7B-Instruct uses bfloat16 inference, temperature 0.7, top-p 0.9 and at most 512 new tokens. Its sampling seed is $z + t + 10{,}000b$ for study seed z, zero-indexed slot t and attempt $b \in \{0, 1\}$. GPT-5.6 Luna uses a text-only bridge at low reasoning effort, tools disabled and the same proposal protocol. It exposes no generation seed; labels identify training/search repetitions. Invalid identifiers, repeats or malformed responses trigger a second attempt; another failure consumes the slot without a fit. Numerically failed fits retain the incumbent and remain in history.

A proposal has exactly four fields:

```
{
  "hypothesis": "why the proposed change may help",
  "experiment": "the intended controlled comparison",
  "expected_effect": "the anticipated metric or diagnostic change",
  "design_id": "D08"
}
```

The descriptive fields are nonempty rationale strings; the registered identifier determines the executed configuration.

Evidence cards contain the configuration, hypothesis, best development score/loss, selected round and acceptance decision. Trajectory summaries record the first and last evaluation rounds (5 and 100 for full fits), their sample-weighted training losses, equal-laboratory development scores/losses and last-minus-first changes in both development metrics. Main-study summaries also include the worst laboratory score and evaluating-laboratory count. Source-transfer and repartitioning studies specify training participation and evaluation panels separately. Cards remain ordered, including unsuccessful trials.

An observed proposal sequence. For Norman with ten laboratories and seed 42, first-slot D06 ties D00's development Top-1 of 40.00 but lowers loss from 158.98 to 158.94 and is accepted. Second-slot D08 retains the residual module and reduces learning rate from 10^{-3} to 10^{-4}. Between rounds 5 and 100, its Top-1 rises from 10.20 to 51.90 and loss falls from 168.12 to 153.85. It is accepted; the next design, D07, reaches only 45.00 and is rejected. Both outcomes remain in history.

Three trajectories follow; scores/losses are multiplied by 100 and slots are one-indexed.

Study	Selected design	Primary	Loss	First direct / loop slot
PTPC, seed 42	D04	66.58	52.12	5 / 5
VCC, seed 43	D00	24.04	160.78	initial baseline
Norman, seed 42	D08	51.90	153.85	3 / 2

PTPC selects stronger weight decay by loss tie-break at equal AP; VCC retains its initial design. Section 4.7 compares matched search trajectories.

A.3 Independent training-allocation control

This study varies training-round allocation over ten fixed configurations and ten laboratories, using seeds 53–55 on proteomics and the three cell settings.

Learning rate	Weight decay	Momentum	Residual module
0.001	0.001	0	no
0.0001	0.001	0	no
0.0001	0.001	0	yes
0.0001	0.01	0.9	no
0.0001	0.01	0.9	yes
0.001	0.001	0.9	yes
0.01	0.001	0	yes
0.01	0.001	0.9	yes
0.01	0.01	0.9	no
0.001	0.01	0	yes

Uniform allocation fits each design for 80 rounds. Trajectory-guided allocation screens each for 20, then promotes two designs from each of three learning-rate groups: the design with the best screening-checkpoint primary score (lower loss breaks ties), and the remaining design with the largest round-5-to-20 score improvement (lower round-20 loss breaks ties). Complete ties follow table order. The six promoted designs restart from initialization for 100 rounds; final selection uses only their full-training checkpoints. The other four stop after screening.

Both schedules use 800 aggregation rounds, $10\times80 = 10\times20+6\times100$, or 8,000 local epochs. Evaluation every five rounds gives 160 aggregate evaluations of ten laboratories each. Training/evaluation counts are matched; the scheduler generates no language-model proposals.

The 12 task–seed held-out comparisons yield four wins, eight ties and no losses. Norman's mean gain is 10.84 Top-1 points: +10.00, +11.70 and +10.81 by seed, with hierarchical paired-bootstrap 95% interval [4.54, 17.73]. Tahoe gains +5.97 in one seed and ties twice; VCC and PTPC tie throughout. A separate TAPB development replay over seven shared designs uses 525 uniform versus 520 trajectory-guided candidate-rounds and gains +0.39 AUROC points (two wins, one tie). It is excluded from the held-out comparison.

A.4 Data construction and task adapters

Drug–target interaction. BindingDB contains 50,149 fitting and 5,604 development pairs. Assigning canonical compounds to ten laboratories gives 4,932–5,058 training and 517–643 development pairs each. Test sets contain 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein pairs. TAPB combines frozen public protein features with trainable drug, fusion and classification components, optionally adding a residual adapter before confounder-averaged prediction. Its objective combines classification NLL and masked-language loss.

Observed-response proteomics. PTPC has 333 compound groups and 491 conditions: 386 fitting (35–42 per laboratory) and 105 development conditions; the test set has 148. Each 6h/24h profile undergoes log1p, within-profile min–max normalization over finite observations, zero filling and unit L2 normalization. The 2,048-bit Morgan vector is L2-normalized and scaled by 0.5. Two 5,171-dimensional protein blocks plus drug features form a 12,390-dimensional float32 input. The ProteinTalks-derived head combines protein/drug convolutional branches and a 64–32–1 classifier (464,193 D00 parameters), trained from random initialization with binary cross-entropy. All methods share these inputs and architecture; selection uses AP.

Cell inputs and response units. VCC, Norman and Tahoe use whole-intervention fitting/development/test splits of 60/20/20, 70/20/27 and 180/60/67. Training uses one mean profile per intervention. VCC development/test queries average disjoint groups of up to 64 cells, omitting a final group smaller than 16. Norman uses released scVI-derived differential-expression profiles per gene pair; Tahoe uses one aggregate profile per compound at the selected cell context and dose.

scDEBART encodes the shared untreated reference and candidate perturbation; its features are frozen while the response head trains. The corrected extractor implements three additive blocking masks, preserving pretrained weights, gene panels, targets, partitions and candidate options. Genetic scoring excludes the union of directly perturbed genes. Tahoe appends a fixed 128-dimensional projection (seed 1234) of a 2,048-bit Morgan fingerprint to 768-dimensional backbone features, zero-initializing

Table A.1: Control comparison for the task-reference models. The fixed and feedback-free direct references train or search on a single site, so they are reported on one laboratory; BioCoLoop is reported at one and ten laboratories so that the effect of laboratory participation can be read under the same research loop. Entries are held-out mean ± sample SD across completed training/search seeds, multiplied by 100. The data partitions remain fixed; this SD measures run-to-run variation, not dataset uncertainty. DTI uses the random held-out endpoint. Bold marks all displayed maxima.

		DTI	Proteomics	Cell perturbation		
Labs	Method	AUROC ↑	AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
1	Fixed model	82.83±1.22	25.05±4.56	29.94±0.00	**25.56**±0.00	19.40±0.00
1	Direct	89.89±1.02	20.13±0.07	21.09±4.38	14.15±0.00	15.42±6.21
1	Our loop	89.89±1.02	20.13±0.07	20.22±2.22	14.15±0.00	15.42±6.21
10	Our loop	**93.76**±1.11	**36.99**±0.22	**31.54**±0.08	22.25±0.45	**22.39**±2.59
Completed seeds ($n/3$)		3/3	3/3	3/3	3/3	3/3

the additional input weights. Training minimizes response MSE plus 0.1 times local-minibatch intervention contrastive loss, with cosine-logit temperature 0.1.

Five-option queries and scoring. Each development/test intervention receives four distinct distractors sampled uniformly without replacement from its *own split*, excluding the true intervention. After adding the true option, positions are shuffled. This roster is fixed across its response queries, training seeds and methods. Split and roster generation use seed 20260918. Training instead draws contrastive negatives from the local minibatch, not these rosters.

Candidates are ranked by cosine similarity between predicted and observed responses, L2-normalized over scoring genes; ties retain shuffled option order. For G interventions, m_g queries per intervention g, and true-option rank $r_{g,j}$ in query j, the primary score is

$$\text{MacroTop1} = \frac{1}{G}\sum_{g=1}^{G}\frac{1}{m_g}\sum_{j=1}^{m_g}\mathbf{1}\{r_{g,j}=1\}.$$

Interventions receive equal weight regardless of query count. Micro Top-1, mean reciprocal rank and Top-3 average over queries. Development loss is cross-entropy over the five cosine scores at temperature 0.1. Selection computes metrics within each laboratory, then averages laboratories equally.

Identification diagnostics. Frozen-scDEBART diagnostics span 7.5–31.6% macro Top-1 (chance: 20%) and 36.00–57.00 MRR (chance: 45.67%, from $(1+1/2+1/3+1/4+1/5)/5$). The untrained public head scores 7.5% on Norman. True-option positions are near-uniform; an oracle constructed from observed held-out response prototypes retrieves 94.5% on VCC and 100% on Norman.

A.5 Complete comparisons and uncertainty

The tables report controls, secondary metrics and all three DTI test sets. Bold marks the best task/metric values, including ties; higher is better except for binary and cell cross-entropy.

Table A.2: Held-out secondary metrics for the task-reference models, averaged over seeds 42–44 and multiplied by 100. Bold indicates all tied maxima, or minima for loss.

Method	Labs	Metric 1	Metric 2	Metric 3	Metric 4
DTI ($n = 3$)		AUROC	AP	MCC	BCE
Fixed model	1	82.83	81.57	49.95	52.37
Direct	1	89.89	87.77	65.71	61.02
Our loop	1	89.89	87.77	65.71	61.02
Our loop	10	**93.76**	**92.73**	**73.71**	**36.59**
Proteomics ($n = 3$)		AP	AUROC	BCE	Acc.
Fixed model	1	25.05	65.49	42.65	**85.14**
Direct	1	20.13	61.80	44.34	81.76
Our loop	1	20.13	61.80	44.34	81.76
Our loop	10	**36.99**	**73.87**	**39.83**	**85.14**
VCC ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	33.61	**56.53**	**73.89**	**154.51**
Direct	1	21.48	47.10	62.78	160.82
Our loop	1	20.74	46.94	64.26	160.86
Our loop	10	**33.98**	56.34	69.44	158.14
Norman ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	**25.56**	48.50	63.93	**158.74**
Direct	1	14.15	42.12	58.35	160.39
Our loop	1	14.15	42.12	58.35	160.39
Our loop	10	22.25	**50.07**	**70.96**	158.83
Tahoe ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	19.40	45.00	58.21	163.08
Direct	1	15.42	42.60	56.72	161.90
Our loop	1	15.42	42.60	56.72	161.90
Our loop	10	**22.39**	**47.95**	**61.19**	**160.75**

Table A.3: All three frozen TAPB held-out endpoints under the committed common protocol. Scores are completed-seed means multiplied by 100; N/A means no complete independently verified evaluation. The main DTI endpoint is random held-out.

Endpoint	Method	Labs	AUROC	AP
Random	Fixed model	1	82.83	81.57
Random	Direct	1	89.89	87.77
Random	Our loop	1	89.89	87.77
Random	Our loop	10	**93.76**	**92.73**
Unseen drug	Fixed model	1	81.96	85.78
Unseen drug	Direct	1	88.39	89.35
Unseen drug	Our loop	1	88.39	89.35
Unseen drug	Our loop	10	**91.65**	**92.58**
Unseen protein	Fixed model	1	76.40	78.75
Unseen protein	Direct	1	83.26	80.53
Unseen protein	Our loop	1	83.26	80.53
Unseen protein	Our loop	10	**87.63**	**85.07**

Three-seed means and sample SDs describe training/search variability on fixed partitions. Sample-level intervals use 1,000 fixed-seed paired cluster resamples of retained predictions: drug identity

for random/unseen-drug DTI, protein sequence for unseen-protein DTI, compounds for proteomics, and interventions for cells. Contrasted methods share resampled clusters. Allocation uses 5,000 hierarchical paired resamples, first drawing training seeds and then compound/intervention groups within each sampled seed.

BioCoLoop Statistical Supplement, Tables S.1–S.2, reports all per-seed differences and intervals for the main contrasts and three DTI test sets, including zero and negative differences. These model-conditional intervals describe test-sample uncertainty; several cell intervals include zero.

Configurations, proposals, development histories and saved predictions support reconstruction; the reproducibility statement details their availability and verification.

A.6 Scope and limitations

Experiments simulate laboratories on shared compute with public data; they do not measure communication/straggler costs or provide formal privacy guarantees. Development is restricted to four registered factors, including the optional residual module. Qwen generation is seed-reproducible; Luna exposes no generation seed. Appendix C documents public-agent adaptations, component restrictions and incomplete executions. Main-table workflows lack a separately compute-matched centralized control. Appendix B's fixed-pool $K = 1$ sensitivity does use all target training data, while changing local optimization and, for cells, the contrastive-negative pool relative to $K = 10$.

Appendix B. Laboratory, source and proposer sensitivity

Sensitivity study design. We vary laboratory participation, data sources, proposer and search budget using the main study's target scorers. Laboratory-count studies cover all five endpoints at $K \in \{1, 2, 5, 10\}$ (DTI: seed 61; proteomics/cell: seeds 61–63). Independent-source DTI/proteomics use seed 61; external-source VCC uses seeds 61–63. Proposal studies compare 12 designs/six slots on all endpoints (DTI: seed 42; others: 61) and 36 designs/24 slots on four lighter endpoints (seed 61), with 100 rounds per candidate. Budget effects are assessed within each protocol because the libraries differ. Main-table seeds 42–44 and protocols remain unchanged.

B.1 Number of laboratories

Increasing participation adds the first K original laboratories and their training/development data. The fixed-pool study instead merges all ten training partitions into K groups while preserving equal-weight evaluation over the original ten development panels. Thus the first experiment adds evidence, whereas the second changes training partitions without changing the selection population. Both use the default design, 100 rounds and the same $K = 10$ fit. For cell tasks, repartitioning also changes within-group contrastive negatives and optimization steps. Zero between-seed SD can arise from identical single-batch updates to a fixed pretrained head; it describes training variability on these fixed partitions.

Table B.1: Increasing participation. A nested set of the original ten client partitions participates, so increasing K also adds training data. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean ± sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	83.65	84.43	89.62	**93.88**
PTPC	AP	26.03 ± 5.57	37.38 ± 0.60	29.63 ± 2.50	**37.54 ± 0.27**
VCC	Macro Top-1	30.02 ± 0.78	**39.56 ± 0.11**	31.88 ± 0.00	31.48 ± 0.16
Norman	Macro Top-1	**25.56 ± 0.00**	18.22 ± 0.00	19.68 ± 1.11	15.48 ± 0.00
Tahoe	Macro Top-1	19.40 ± 0.00	25.37 ± 0.00	20.90 ± 0.00	**26.37 ± 0.86**

Table B.2: Fixed data pool. All target data are retained and merged into K clients; only the partition count changes. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean ± sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	85.47	88.96	88.05	**93.88**
PTPC	AP	34.59 ± 5.66	37.36 ± 1.89	**39.19 ± 0.43**	37.54 ± 0.27
VCC	Macro Top-1	29.49 ± 0.00	28.80 ± 0.20	31.06 ± 0.22	**31.48 ± 0.16**
Norman	Macro Top-1	**49.63 ± 4.20**	19.56 ± 0.20	26.12 ± 0.11	15.48 ± 0.00
Tahoe	Macro Top-1	22.39 ± 0.00	21.39 ± 0.86	20.40 ± 0.86	**26.37 ± 0.86**

B.2 Independent sources and target evaluation

For DTI and cell transfer, the full target training pool and each auxiliary collection form separate laboratories, combined by sample-weighted aggregation (Table B.3). Target-only, individual-source and combined-source arms share initialization, original target-development panels for checkpoint selection, and the target test set.

BioSNAP, Davis and Human from the pinned TAPB release contribute 19,092, 1,721 and 2,929 DTI pairs after de-duplication against target splits and preceding sources; all five source arms use 100 rounds. VCC adds Replogle K562, Nadig HepG2 and Jiang IFNB/K562, retaining 60, 60 and 58 interventions (Replogle et al., 2022; Nadig et al., 2025; Jiang et al., 2025a). Source panels retain measured genes and exclude target development/test intervention identities from training; the source manifest records subsets, feature intersections and exclusions.

Table B.4 retains every source-specific gain and decrease. Across three cell-task seeds, Jiang raises Top-1 by 3.58 points (paired intervention-bootstrap 95% interval: $[-2.11, 9.51]$).

Table B.3: Source and context mapping in the cross-source experiments. The DTI units are database collections, the cell units preserve their study and biological scenario, and each proteomic auxiliary study owns one client and its response head. The ten PTPC clients remain partitions of one target corpus. Biological contexts within a study are not counted as additional independent studies.

Task	Client unit	Sources and biological contexts
DTI	Four data collections	Target BindingDB; BioSNAP; Davis; Human. Each collection contributes one client after pair-level de-duplication.
Cell	Four study/scenario clients	Target VCC; Replogle K562; Nadig HepG2; Jiang IFNB/K562. Each auxiliary source contributes one client.
Proteomics	Ten target partitions plus three source clients	Clients 0–9: PTPC partitions; client 10: Lin (A549, H292, PC9, PSC1); client 11: Ruprecht (A549, Calu6, Calu1, 2030, 2122); client 12: decryptE (Jurkat, five doses).

Table B.4: Independent-source transfer on the same target heldout set, including every prespecified source and their combination. DTI (seed 61) adds BioSNAP, Davis or Human; VCC (seeds 61–63, mean $\pm$ SD) adds Replogle, Nadig or Jiang. The PTPC auxiliary multi-task model (seed 61) adds Lin or Ruprecht and is compared only with its matched auxiliary target-only model. The two-source PTPC protocol shown here uses Lin and Ruprecht; the three-source extension is reported separately in Table B.6. Changes are paired against target-only training; bold marks the best score within a target/model block. Scores are multiplied by 100.

Target	Additional source	Metric	Score ↑	Δ (pp)
DTI	Target only	AUROC	85.47	+0.00
DTI	+ BioSNAP	AUROC	82.44	-3.03
DTI	+ Davis	AUROC	87.86	+2.39
DTI	+ Human	AUROC	**87.95**	+2.48
DTI	+ All three	AUROC	84.30	-1.18
VCC	Target only	Macro Top-1	29.49 $\pm$ 0.00	+0.00 $\pm$ 0.00
VCC	+ Replogle	Macro Top-1	27.12 $\pm$ 0.08	-2.37 $\pm$ 0.08
VCC	+ Nadig	Macro Top-1	26.94 $\pm$ 0.18	-2.55 $\pm$ 0.18
VCC	+ Jiang	Macro Top-1	**33.07 $\pm$ 0.00**	+3.58 $\pm$ 0.00
VCC	+ All three	Macro Top-1	27.33 $\pm$ 0.00	-2.15 $\pm$ 0.00
PTPC auxiliary	Target only	AP	34.98	+0.00
PTPC auxiliary	+ Lin	AP	35.01	+0.03
PTPC auxiliary	+ Ruprecht	AP	**35.50**	+0.52
PTPC auxiliary	+ Both	AP	35.31	+0.34

Table B.5: Fixed-design source–scenario ablation on VCC. K: training clients; N: training intervention conditions. The two $K = 4, N = 60$ arms share a 15-condition VCC anchor and differ in the sources of the other three clients. All arms use 100 rounds and the same target evaluation. Mean $\pm$ sample SD over three seeds; best means are bold. Cell counts and gene panels remain source-specific; protocol details are in Appendix B.

Training sources	K	N	Top-1 (%) ↑	MRR (%) ↑
VCC, limited data	1	15	29.36 ± 0.26	$\mathbf{56.97 \pm 0.19}$
VCC, all target data	1	60	29.43 ± 0.09	56.00 ± 0.09
VCC, same-source split	4	60	$\mathbf{30.70 \pm 0.42}$	56.24 ± 0.04
VCC + external studies	4	60	28.82 ± 0.00	56.30 ± 0.00

Matched source–scenario laboratories. Table B.5 compares 15 or 60 VCC interventions in one laboratory, the same 60 split across four laboratories, and four source-defined laboratories with 15 interventions each (VCC, Replogle/K562, Nadig/HepG2, Jiang/IFNB/K562). Both four-laboratory arms share a 15-intervention VCC anchor. Hash-based auxiliary sampling excludes target training/development/test identities and cross-source duplicates before fitting. Intervention counts match, but cell counts differ: 77,771 same-source versus 33,469 cross-scenario cells; source-specific gene panels and loss masks are retained. All arms use identical head initialization, the default design, 100 rounds, seeds 61–63 and no proposals. The original ten target-development panels select checkpoints; testing uses 20 interventions and 360 five-option queries. Independent rescoring verifies all twelve fits. Cross-scenario Top-1 is lower in every seed (mean difference -1.88 points); MRR differs little.

Two-source proteomic transfer and proposal search. Lin (Lin et al., 2026) and Ruprecht (Ruprecht et al., 2020) each add one study-level laboratory to the ten target partitions. Their continuous responses differ from the binary target endpoint. A shared masked-protein encoder of width 64 and drug encoder of width 32 therefore feed separate heads for target binary efficacy, Lin CellTiter-Glo response and Ruprecht potency intervals. Encoders use sample-weighted aggregation; each private head is updated only by its own source laboratories. This auxiliary adapter has its own target-only control, separate from the main PTPC predictor.

Lin provides 28 conditions (20 training/eight development, from five training and two development drugs), with 24-hour, 10-μM protein log-fold changes and the mean of three released CellTiter-Glo ratios; the HCT16/HCT116 alias is excluded from auxiliary training. Ruprecht provides 109 conditions (85 training/24 development, from 17 training and five development drugs), pairing 24-hour protein log-fold changes with $\log_{10}$-nM EC50: 49 exact, 18 approximate and 42 right-censored labels. The loss is squared distance to the potency interval; explicit lower bounds or values $\geq 10{,}000$ nM are censored, and approximate values receive factor-of-two intervals. Profiling dose is omitted from model inputs, although source acquisition doses relate to response. The target has 386 training and 105 development examples, with 148 test observations across 92 compounds and 22 positives.

Each source setting uses 100 rounds and an independent two-proposal Luna search. All 12 fits and eight valid calls complete; Fixed and Loop have identical rankings in all four settings. Their AP/AUROC are 34.98/74.75 (target only), 35.01/74.78 (+Lin), 35.50/74.28 (+Ruprecht) and 35.31/74.17 (+both). Paired compound-bootstrap 95% intervals (1,000 resamples) for the three AP changes are $[-0.20, 0.28]$, $[-1.46, 2.64]$ and $[-1.62, 2.49]$ points. These are source-transfer results without an additional history-search gain.

Three-source proteomic transfer. decryptE (Eckert et al., 2025) contributes Jurkat proteomes after 18 hours at 1, 10, 100, 1,000 and 10,000 nM, predicting continuous metabolic activity relative to DMSO. Inputs combine log-transformed normalized protein intensities, an observation mask, drug fingerprints and dose indicators. Compounds are aligned by published names/aliases and structures; proteins by UniProt identifier. Splits keep all doses of a chemical graph together, with training-only normalization and target development/test exclusions by canonical and stereo-agnostic identity. Of 145 processed matrices, one is an additional Vincristine copy, leaving 144 compounds. Excluding 12 target overlaps, six unsupported names and two identity/structure conflicts yields 124 compounds/620 conditions: 495 training observations from 99 compounds and 125 development observations from 25 compounds. The source manifest preserves exclusions and annotations.

All four arms—target only, +decryptE, +Lin/Ruprecht and +all three—share a four-head model, seed-61 initialization, optimizer, sample-weighted shared-encoder aggregation and 100 rounds. Each source owns its private head. Equal-laboratory target-development AP selects checkpoints, breaking ties by binary cross-entropy; auxiliary development scores are diagnostic. These fixed-design arms use zero proposals.

Table B.6: Three-source proteomic transfer on PTPC. All four arms use the same four-head adapter, seed 61, 100-round fitting schedule and target-development checkpoint rule. AP and AUROC are multiplied by 100; bold identifies the best value in each metric. The target test set contains 148 observations from 92 compounds. These are fixed-design source comparisons; the separate two-source proposal pilot is described in Appendix B.2.

Additional source	AP ↑	AUROC ↑	Δ AP (pp)
Target only	34.98	**74.75**	+0.00
+ decryptE	35.03	74.21	+0.05
+ Lin + Ruprecht	35.31	74.17	+0.34
+ Lin + Ruprecht + decryptE	**35.68**	74.28	+0.70

All three sources raise AP from 34.98 to 35.68 (+0.70 points); adding decryptE to Lin/Ruprecht adds 0.37 AP and 0.11 AUROC points. Target-only retains the highest AUROC (74.75 versus 74.28). The four single-seed CPU fits take 101.21 seconds; independent rescoring of sealed predictions agrees exactly.

One study per training laboratory. Merging the 386 target training examples into one laboratory gives $K = 1, 2, 3, 4$ study-defined laboratories with 386, 881, 491 and 986 conditions, respectively. Lin, Ruprecht and decryptE each remain a separate laboratory. The four-head model, seed-61 initialization, losses, aggregation and private-head ownership are unchanged. The target now executes seven minibatches per local epoch rather than aggregating ten partitions, requiring a separately trained target-only control. All four arms use 100 rounds, zero proposals and equal-weight checkpoint selection over the original ten target-development panels. Training participation therefore changes while the target evaluation population stays fixed. The CPU run takes 76.18 seconds; independently rescored AP/AUROC differ by at most 1.11×10^{-16}.

All three auxiliaries raise AP from 34.41 to 37.53 and AUROC from 72.33 to 74.39; decryptE adds 1.44 AP points beyond Lin/Ruprecht. These fixed-design, single-seed comparisons vary training data and participating studies.

Within-study proteomic contexts. HCC1143, HCC1806 and MDA-MB-453-1 are three contexts within mtPTDS (Sun et al., 2026), each with 47 conditions and 3,501 target-overlapping proteins. Training excludes target development/test compounds. These runs use the original proteomic adapter and seeds 61–63.

B.3 Research-model and six-slot proposal comparison

The 12-design study (Appendix A) gives direct search untried identifiers without scores; the loop additionally receives development history. Both share initialization and the first proposal. Each slot allows two attempts at the four-field schema; failures consume budget and remain recorded. Valid candidates restart from the common task/seed initialization for 100 rounds. Prefixes 0, 1, 2, 4 and 6 are scored after search; identical cached fits retain their logical budget charge. Both backends share task messages, designs, training and proposal budgets. Qwen uses Appendix A's sampling settings; `gpt-5.6-luna` uses a text-only, tool-disabled service with retained requests/responses. Its decoding controls and parameter count are not exposed.

Across ten endpoint/backend pairs, Norman/Luna gains 0.81 Top-1 points, DTI/Luna loses 0.33 AUROC points and eight tie. Independent rescoring verifies all twenty DTI prefix scores on 5,505 test pairs and their checkpoint/output hashes. Figure 4 shows every prespecified development/test prefix.

B.4 Extended search and task-dependent saturation

Both backends and modes run 24 slots on PTPC, VCC, Norman and Tahoe using 36 designs: three learning rates, three weight decays, two coordinator-momentum values and two residual-head set-

Table B.7: One worker per proteomic study. The target training pool is merged into one client; Lin, Ruprecht and decryptE each form one additional client. K counts training clients and N counts training conditions. All four fixed-design arms use 100 rounds, the same initialization (seed 61), and equal-weight evaluation over the original ten target development panels. AP and AUROC are percentages on the common target held-out set; best values are bold.

Training sources	K	N	AP ↑	AUROC ↑
Target study	1	386	34.41	72.33
+ decryptE	2	881	36.58	73.81
+ Lin + Ruprecht	3	491	36.09	72.69
+ Lin + Ruprecht + decryptE	4	986	**37.53**	**74.39**

Table B.8: Within-study PTPC context transfer, reported separately from independent-source transfer. HCC1143, HCC1806 and MDA-MB-453-1 belong to the same mtPTDS study. Values are mean ± sample SD over seeds 61–63, multiplied by 100. Confidence intervals resample target compounds while retaining paired seed predictions. These contexts are not additional independent studies; the original model and results are unchanged.

Additional context	AP ↑	Δ AP (pp)	Paired 95% CI
Target only	34.59 ± 5.66	+0.00 ± 0.00	[0.00, 0.00]
+ HCC1143	34.94 ± 5.15	+0.35 ± 0.56	[-1.22, 1.60]
+ HCC1806	35.23 ± 5.35	+0.64 ± 0.32	[-0.78, 1.77]
+ MDA-MB-453-1	35.20 ± 5.31	+0.61 ± 0.44	[-0.85, 2.04]
+ All three	**35.47** ± **6.36**	+0.87 ± 8.89	[-2.44, 3.99]

tings. Prefixes are 0, 1, 2, 4, 6, 8, 12, 16, 20 and 24. Initialization, fitting, selection and invalid-slot accounting follow B.3; DTI remains a six-slot comparison.

Luna returns valid designs in 192/192 slots and Qwen in 116/192. Of 16 trajectories, twelve have no primary-score improvement in the final eight slots, and ten have no score/loss selection change. Each still has 11–24 unvisited designs: the observed plateaus precede library exhaustion. Table B.10 records first attainment of the lower terminal development score in each direct/loop pair, a target reached by both runs. This retrospective target is not a stopping rule: all 24 slots ran, so earlier crossing is not measured compute saving. Feedback reaches Norman/Luna's target in 4 versus 22 slots; direct reaches Tahoe/Luna's in 5 versus 13. Proteomics retains the initial design. Norman's retained proposals (Table B.12) first lower the learning rate to 10^{-4}, then remove the residual head to reach the final primary target; later loss-tiebreak selections may change the model without increasing Top-1. Meanwhile, Qwen's VCC test Top-1 falls from 31.50 at slot 6 to 19.61 at slot 24 as development selection improves; both curves are retained below.

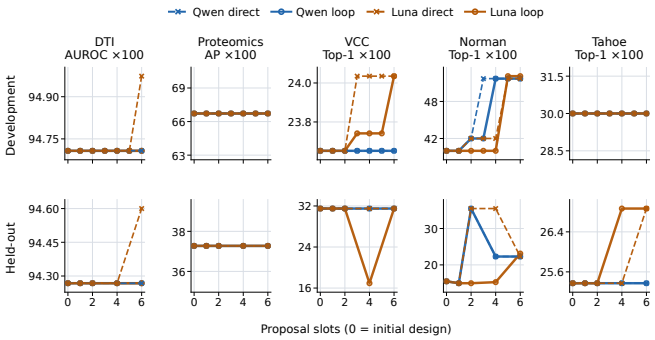


Figure 4: Research trajectories with and without evidence history in the six-slot protocol. Curves report development selection and held-out scores at each proposal prefix, where each candidate receives 100 rounds and slot 0 is the initial design.

Table B.9: Original and extended proposal searches, shown together without mixing configuration libraries. D and H denote direct and history-guided search, and @ denotes the number of proposal slots. Every candidate receives 100 training rounds using the same ten laboratories. Panel (a) uses DTI seed 42 and other-task seed 61; panel (b) uses seed 61. The six-slot prefixes in the two panels belong to different searches. DTI uses AUROC, proteomics uses average precision, and cell tasks use Top-1, all multiplied by 100. Bold denotes the highest displayed score in each row, including ties; dashes indicate runs not performed.

Task	Proposer	Fixed	D@6	H@6	D@24	H@24
(a) 12 designs, six proposal slots						
DTI	Luna	94.27	**94.60**	94.27	–	–
DTI	Qwen	**94.27**	**94.27**	**94.27**	–	–
Proteomics	Luna	**37.27**	**37.27**	**37.27**	–	–
Proteomics	Qwen	**37.27**	**37.27**	**37.27**	–	–
VCC	Luna	**31.50**	**31.50**	**31.50**	–	–
VCC	Qwen	**31.50**	**31.50**	**31.50**	–	–
Norman	Luna	15.48	22.30	**23.11**	–	–
Norman	Qwen	15.48	**22.30**	**22.30**	–	–
Tahoe	Luna	25.37	**26.87**	**26.87**	–	–
Tahoe	Qwen	**25.37**	**25.37**	**25.37**	–	–
(b) 36 designs, 24 proposal slots						
Proteomics	Luna	**37.27**	**37.27**	**37.27**	**37.27**	**37.27**
Proteomics	Qwen	**37.27**	**37.27**	**37.27**	**37.27**	**37.27**
VCC	Luna	**31.50**	16.94	16.94	20.23	20.01
VCC	Qwen	**31.50**	**31.50**	**31.50**	19.61	19.61
Norman	Luna	15.48	**35.48**	23.11	23.11	23.11
Norman	Qwen	15.48	**22.30**	**22.30**	**22.30**	**22.30**
Tahoe	Luna	25.37	13.43	**26.87**	13.43	13.43
Tahoe	Qwen	25.37	**26.87**	**26.87**	13.43	13.43

Table B.10: Retrospective development-trajectory diagnostics for the two proposal-search protocols. Each D/H pair gives direct/history. The target is the lower terminal development score of the two runs; First is its earliest attainment, with slot 0 denoting initialization. Last gain records the final strict primary-score increase; Last retain also includes loss-tiebreak selections. All six or 24 slots were executed: these target crossings are descriptive and did not save training budget. Valid counts report accepted proposal formats, not retained model improvements.

Task	Proposer	Dev. target	First D/H	Last gain D/H	Last retain D/H	Valid D/H
(a) 12 designs, six proposal slots						
DTI	Luna	94.71	0/0	6/0	6/0	6/6
DTI	Qwen	94.71	0/0	0/0	0/0	6/6
Proteomics	Luna	66.72	0/0	0/0	0/0	6/6
Proteomics	Qwen	66.72	0/0	0/0	0/0	6/5
VCC	Luna	24.04	3/6	3/6	3/6	6/6
VCC	Qwen	23.65	0/0	0/0	0/0	6/6
Norman	Luna	51.70	5/5	5/5	5/5	6/6
Norman	Qwen	51.70	3/4	3/4	3/4	6/6
Tahoe	Luna	30.00	0/0	0/0	6/3	6/6
Tahoe	Qwen	30.00	0/0	0/0	0/0	4/4
(b) 36 designs, 24 proposal slots						
Proteomics	Luna	66.72	0/0	0/0	0/0	24/24
Proteomics	Qwen	66.72	0/0	0/0	0/0	19/14
VCC	Luna	27.68	16/9	16/10	16/10	24/24
VCC	Qwen	24.83	18/23	18/23	18/23	11/16
Norman	Luna	52.10	22/4	22/4	22/7	24/24
Norman	Qwen	51.70	2/5	2/5	22/19	15/12
Tahoe	Luna	35.00	5/13	5/13	5/13	24/24
Tahoe	Qwen	35.00	18/14	18/14	18/14	14/15

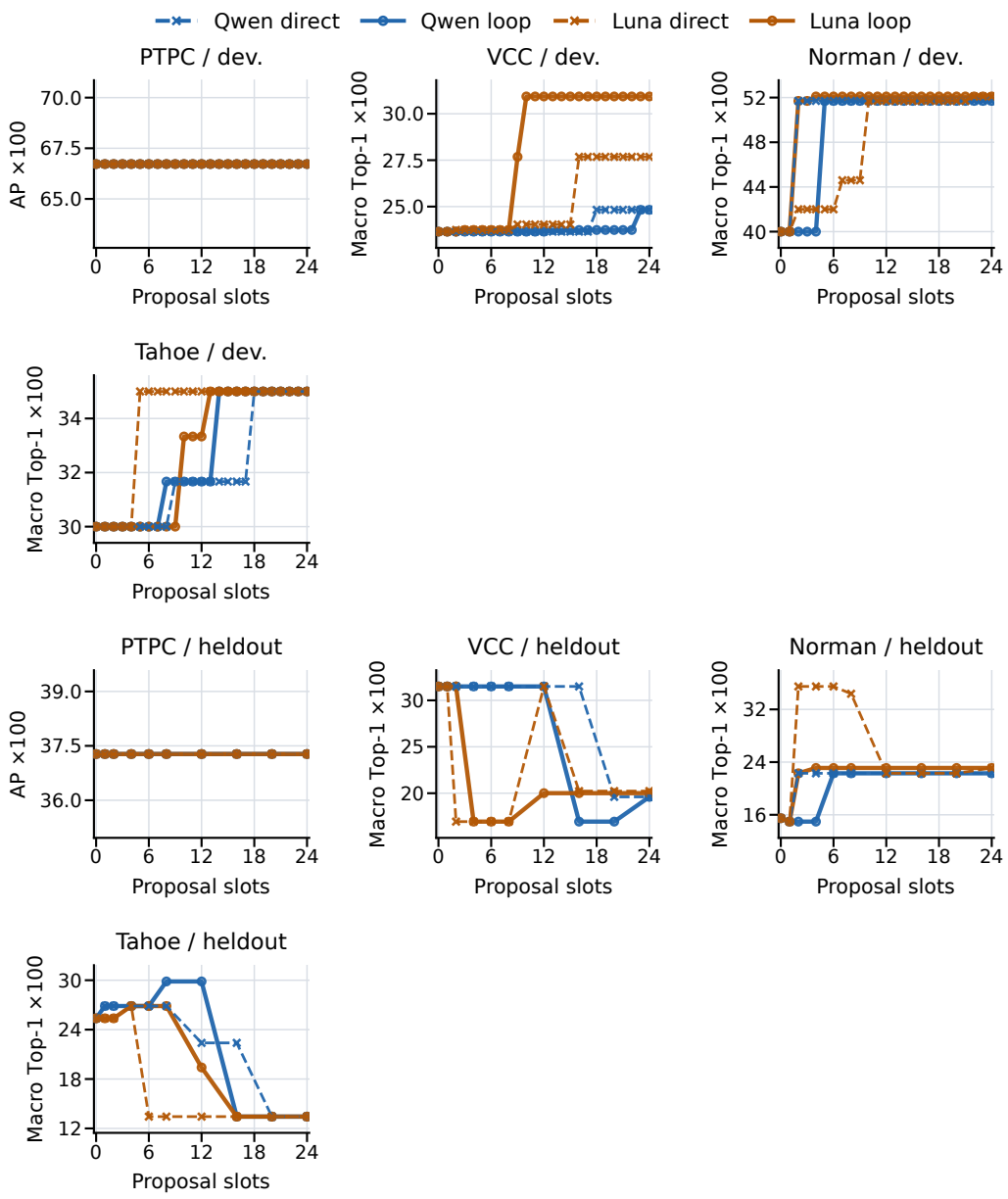


Figure 5: Development selection (upper panels) and held-out performance (lower panels) in the separate 36-design, 24-slot protocol on four endpoints. Curves are paired within the same backend, initialization and design space. Score plateaus may still include loss-tiebreak selections. All prespecified prefixes are displayed. Six-slot points here belong to this 36-design protocol, not the separate 12-design experiment.

B.5 Executed designs and proposal traces

The short study uses D00–D11 (Appendix A.2); the extended study uses all $3 \times 3 \times 2 \times 2 = 36$ combinations below. Identifier LiWjMkRl indexes each factor ($i, j \in \{0, 1, 2\}$; $k, l \in \{0, 1\}$). Thus LOW1MOR1 means learning rate 10^{-4}, weight decay 10^{-3}, zero coordinator momentum and an enabled residual head. Proximal coefficients are zero; task interfaces stay fixed.

Table B.11: Exact factor-to-index mapping for the extended 36-design library. Identifiers concatenate the four prefix–index pairs in L, W, M, R order, and all combinations of the listed values are included.

Factor	Identifier prefix	Index 0	Index 1	Index 2
Learning rate	L	10^{-4}	10^{-3}	10^{-2}
Weight decay	W	0	10^{-3}	10^{-2}
Coordinator momentum	M	0	0.9	–
Residual module	R	disabled	enabled	–

The reproducibility archive retains all 504 attempted slots (120 short, 384 extended), including acceptance decisions, configurations, incumbent-relative changes and measured development scores.

Table B.12: All retained proposals for the Norman/Luna extended 36-design, 24-slot study, seed 61. The two modes share initialization and first proposal. lr, wd, m and R denote learning rate, weight decay, server momentum and residual-head indicator. Development selection is followed by evaluation at every prespecified heldout prefix; the full attempted-slot trace is retained separately.

Mode	Slot	Executed configuration	Dev. Top-1
Direct	0	Common initial recipe	40.00
Direct	1	lr=0.001, wd=0.001, m=0, R=1	40.00
Direct	2	lr=0.001, wd=0.001, m=0.9, R=1	42.00
Direct	7	lr=0.001, wd=0.01, m=0.9, R=1	44.60
Direct	10	lr=0.0001, wd=0, m=0, R=1	51.70
Direct	11	lr=0.0001, wd=0.001, m=0, R=1	51.70
Direct	22	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	0	Common initial recipe	40.00
Loop	1	lr=0.001, wd=0.001, m=0, R=1	40.00
Loop	2	lr=0.0001, wd=0.001, m=0, R=1	51.70
Loop	4	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	7	lr=0.0001, wd=0, m=0, R=0	52.10

Appendix C. Public self-improving-agent comparison

C.1 Shared biological task interface

AI-Scientist-v2 (Yamada et al., 2025) and AI-Researcher (Tang et al., 2025) use BioCoLoop's fitting and evaluation service, at source versions `96bd51617cfd` and `f9a6f8480860`, respectively (SakanaAI, 2025; HKUDS, 2025). Both use the registered executable design space of twelve configurations, at most six proposals, 100 rounds per valid candidate and the shared held-out scorer. The service fits D00 afresh, then executes submitted `design_id` values; the hypothesis, experiment and expected-effect fields record rationale. Repeated or invalid requests consume slots, and agents may stop early. The original development score/loss rule selects checkpoints.

AI-Scientist-v2 retains its experiment manager, tree-search paths, reviews and journal. AI-Researcher retains native idea selection, planning, implementation, review handoffs and iterative analysis. Prompts use the registered design interface; arbitrary generated-code execution, external retrieval, figure review and manuscript generation are disabled. The matched blocks use local Qwen2.5-7B-Instruct (temperature 0.7, top-p 0.9, at most 512 generated tokens per call) and GPT-5.6 Luna (low reasoning effort, text-only bridge, tools disabled). Adapters allow at most 160 model calls and 28,000 input tokens per call, raised to 65,536 for the four starred Luna AI-Researcher retries (C.2). Qwen uses generation seeds; Luna labels identify training/search repetitions. Per-run call counts, token usage and transport receipts document the variable language-model costs.

C.2 Prediction quality and execution outcomes

A scored run completes its agent's research process within the registered cap, seals development selection and passes independent held-out rescoring. Table 1 summarizes seeds 42–44; fixed, direct and BioCoLoop rows are complete. Daggers mark partial cells and dashes mark unscored cells, without imputation. Partial cells are Qwen2.5-7B-Instruct AI-Researcher on VCC ($2/3$ seeds, F_{tool}) and GPT-5.6 Luna AI-Scientist-v2 on Tahoe ($2/3$, F_{svc}).

Four Luna AI-Researcher cells exceeded the 28,000-token admission (F_{ctx}) and were re-run at 65,536 tokens with the same tasks, seeds, design space, six-proposal cap, training and scorer. Their completed cells carry stars, with per-seed records retained. BindingDB (DTI), the longest native refinement trajectory, remains unscored ($0/3$): one frozen seed reached 29,167 input tokens (F_{ctx}), and two terminated in service transport (F_{svc}). Extended-admission retries also ended in transport before held-out selection was sealed. The revised table reports laboratory access instead of averaging ranks over different completed endpoint sets.

Receipts distinguish native pipeline (F_{pipe}), tool-transport (F_{tool}), registered context-limit (F_{ctx}) and Luna service-transport (F_{svc}) terminations before sealed selection. An earlier seed-42 Qwen AI-Scientist-v2 DTI attempt trained three candidates before substage schema validation failed. That historical attempt is distinct from the completed run represented in the table. Execution failures describe this adaptation and infrastructure, not predictive scores or evidence that the original framework cannot solve the task.

Held-out scoring follows sealed development selection. An independent verifier recomputes each scored cell's metric from saved predictions and checks checkpoint and artifact hashes. DTI discrepancies are at most 10^{-15}. Records retain raw calls, native stage history, candidate requests, selections and verification receipts.

C.3 Task-transport repair

The AI-Researcher adaptation removes repeated task packets from stage prompts, preserving the complete task and evidence in system prompts, native reasoning and tool results. The bridge accepts an unambiguous `parameters` alias for `arguments` and standard single-tool-call envelopes without inferring actions or values. This revision preserves model weights, context allowance, training and scoring. It applies across all five endpoints and the Luna sensitivity; failures remain recorded.

Appendix D. Single-laboratory configuration-search controls

D.1 Search protocol and primary results

We add conventional configuration-search controls to test whether more extensive search within one laboratory can match the collaborative framework. All new controls access only laboratory 0's training and development partitions. They use the same twelve registered designs, task-specific initialization, original GPU training implementation and held-out scorers as the main study. Each design is fitted afresh for 100 rounds at seeds 42–44. Development score, followed by development loss, selects the design and checkpoint.

Random search evaluates D00 and six distinct randomly sampled designs. Bayesian optimization evaluates D00, the first two proposals from the corresponding seeded random-search sequence, and four proposals selected by Gaussian-process expected improvement. Its fixed Matérn-$5/2$ kernel uses unit length scales, diagonal regularization $\alpha = 10^{-6}$, normalized outcomes and an improvement offset of 0.01. Design features encode log-scaled learning rate, ordinal weight decay, normalized momentum and the binary residual flag. Both methods have a nominal budget of 700 candidate-rounds. Exhaustive enumeration evaluates all twelve registered configurations, using 1,200 candidate-rounds; this covers the registered menu rather than a full Cartesian hyperparameter grid. The search decisions are replayed using only the development results of queried configurations from the freshly trained, fixed-seed candidate cache. The BioCoLoop rows reuse the original Qwen2.5-7B-Instruct runs, with and without collaborative participation.

Table D.1: **Single-laboratory search controls.** Mean ± sample SD over three training seeds, in percentage units. Search controls use only lab 0; the history-enabled BioCoLoop references use Qwen2.5-7B-Instruct with one or ten laboratories. The ten-laboratory setting adds training and evaluation evidence. Bold marks the highest displayed mean per column.

Method	Labs	DTI	Proteomics	Cell perturbation		
		AUROC↑	AP↑	VCC Top-1↑	Norman Top-1↑	Tahoe Top-1↑
LLM-free controls						
Fixed model	1	82.83 ±1.22	25.05 ±4.56	29.94 ±0.00	**25.56** ±0.00	19.40 ±0.00
Random@6	1	90.10 ±0.71	20.13 ±0.06	21.10 ±4.41	14.10 ±0.04	17.91 ±3.95
Bayesian@6	1	90.10 ±0.71	20.01 ±0.24	18.61 ±2.78	14.12 ±0.04	16.92 ±4.80
12-design menu	1	90.10 ±0.71	20.13 ±0.06	18.61 ±2.78	14.12 ±0.04	15.42 ±6.21
History-enabled BioCoLoop references						
BioCoLoop	1	89.89 ±1.02	20.13 ±0.07	20.22 ±2.22	14.15 ±0.00	15.42 ±6.21
BioCoLoop	10	**93.76** ±1.11	**36.99** ±0.22	**31.54** ±0.08	22.25 ±0.45	**22.39** ±2.59

The ten-laboratory framework exceeds all three single-laboratory search strategies in mean score on all five endpoints. Including the fixed-model control, it leads on four endpoints; Norman's fixed model retains the best Top-1. This is an access-scenario comparison of complete workflows. Appendix A and Section 4.7 separately hold laboratory access fixed.

D.2 Search variability and held-out uncertainty

Table D.2 integrates random choices over all $\binom{11}{6} = 462$ subsets per training seed, always including D00. Ties integrate proposal orders under the original selection rule. These are expected selected-model scores, not ensembled predictions. The collaborative framework also exceeds these expectations on all five endpoints.

For these additional controls, a separate paired bootstrap uses 10,000 resamples of compounds for proteomics and interventions for the cell tasks. Each resample is shared across methods and the three trained models, and differences are averaged across seeds. These intervals quantify test-group uncertainty conditional on the trained models, complementing the seed standard deviations. The BioCoLoop-minus-random-search 95% intervals are [1.91, 29.97] AP points for proteomics and $[-0.48, 22.49]$, $[-9.53, 26.47]$ and $[-3.98, 12.94]$ Top-1 points for VCC, Norman and Tahoe. The cell intervals span zero despite positive mean differences. These are descriptive intervals without

Table D.2: **Integrating random-search variability.** Expected Random@6 score over all $\binom{11}{6} = 462$ proposal subsets, integrating proposal-order ties. This averages selected-model metrics, not their predictions. Scores are percentages; the final row is the mean ± sample SD across three trained seeds after search randomness is integrated out.

Training seed	DTI AUROC↑	Proteomics AP↑	VCC Top-1↑	Norman Top-1↑	Tahoe Top-1↑
42	88.67	20.42	22.53	15.35	14.38
43	90.08	20.01	19.41	15.33	12.74
44	89.53	20.91	20.82	15.35	19.68
Mean ± SD	89.43 ±0.71	20.44 ±0.45	20.92 ±1.56	15.34 ±0.01	15.60 ±3.63

multiplicity adjustment; the original study's uncertainty protocol remains as specified in Appendix A.5.

D.3 Generalization to unseen DTI entities

The same development-selected checkpoints are evaluated on the original random, unseen-drug and unseen-protein test sets. Exact input-identity checks confirm that none contains a training drug–protein pair; the unseen-drug and unseen-protein sets also have no exact drug-string or protein-sequence overlap with training, respectively. Scores are independently recomputed from saved, row-identified predictions.

Table D.3: **Single-laboratory search across three drug–target test splits.** AUROC and average precision (AP), mean ± sample SD over the same three seeds, expressed as percentages. Each method's development-selected predictor is reused across all three test splits. Lab access, candidate budgets and the historical Qwen BioCoLoop references are as in Table D.1. Bold marks the highest displayed mean per column.

Method	Labs	Random split		Unseen drug		Unseen protein	
		AUROC↑	AP↑	AUROC↑	AP↑	AUROC↑	AP↑
Fixed model	1	82.83 ±1.22	81.57 ±1.54	81.96 ±1.55	85.78 ±1.41	76.40 ±5.85	78.75 ±2.04
Random@6	1	90.10 ±0.71	88.06 ±1.38	89.07 ±0.81	90.19 ±1.38	83.70 ±2.12	81.60 ±4.27
Bayesian@6	1	90.10 ±0.71	88.06 ±1.38	89.07 ±0.81	90.19 ±1.38	83.70 ±2.12	81.60 ±4.27
12-design menu	1	90.10 ±0.71	88.06 ±1.38	89.07 ±0.81	90.19 ±1.38	83.70 ±2.12	81.60 ±4.27
BioCoLoop	1	89.89 ±1.02	87.77 ±1.79	88.39 ±1.66	89.35 ±2.34	83.26 ±2.61	80.53 ±5.10
BioCoLoop	10	**93.76** ±1.11	**92.73** ±1.81	**91.65** ±1.64	**92.58** ±2.16	**87.63** ±2.50	**85.07** ±5.34

BioCoLoop attains the highest mean AUROC and AP on all three DTI test sets. Compared with the three search controls, its AUROC gains are 3.66 points on the random split, 2.57 on unseen drugs and 3.93 on unseen proteins. The benefit therefore extends beyond the primary random-split endpoint.

The additional study comprises 180 complete candidate fits and 18,000 candidate-rounds. Their recorded fit times sum to 13.02 GPU-hours, excluding feature preparation, held-out scoring and statistical analysis. This is the cost of constructing the shared evaluation cache, rather than the cost of a single search. Aggregate per-seed scores, search definitions and source hashes accompany the tables in `tables/single_lab_search/`.

Appendix E. Search controls, allocation and validation

E.1 What the controlled comparison measures

Unlike the ten-versus-one-laboratory access scenario in Table 1, the same-access comparison gives every method ten training laboratories and the same development panels, configuration library, initialization, checkpoint rule and proposal/fit budget. Random search and Bayesian optimization use the same collaborative fitting service without a language-model researcher, isolating configuration search from access to additional measurements.

All controls reuse the complete twelve-design, 100-round caches from Appendix B.3 and the original Qwen/Luna trajectories: DTI uses seed 42 and the other endpoints seed 61. Retrospective cache reuse preserves each query's outcome and logical training charge; it adds search policies, not independent training seeds. This study is separate from the three-seed access comparison.

E.2 Random-search distribution and conventional optimization

Random configuration search provides a direct control for the value of adaptive proposal generation (Bergstra & Bengio, 2012). The fixed model is a one-fit reference; the search controls below receive the same candidate budget as the research loop, rather than that smaller reference budget.

Random search always fits the initial design D00, then selects six distinct configurations from the eleven remaining designs. Enumerating the $\binom{11}{6} = 462$ subsets integrates proposal randomness exactly conditional on the trained candidate cache. Full score/loss ties retain the incumbent; equally ranked newly encountered candidates are averaged over their possible first-encounter orders. A second random control fixes the first proposal to the relevant language-model trajectory and enumerates the remaining $\binom{10}{5} = 252$ subsets. The shared first Qwen/Tahoe slot was invalid: its sensitivity preserves that failed slot and samples five of all eleven remaining designs, giving 462 subsets instead. This distinguishes the shared first proposal from subsequent feedback.

The Bayesian control uses Appendix D's fixed GP–EI recipe and observes only queried development outcomes. Design selections and selection probabilities are fixed before new held-out scoring. Random means average selected-model scores, not predictions; their standard deviations describe configuration-choice variability on one trained cache, not biological or training-seed uncertainty.

All searches receive six slots and a 700-round cap including D00. Invalid slots reduce Qwen history's fitted cost to 600 rounds on proteomics and both Qwen modes to 500 on Tahoe. Matching these realized counts gives Tahoe random expectation 26.30 versus Qwen's 25.37. Fixing a valid shared first proposal gives DTI random expectation 94.43 versus history's 94.27. These sensitivities, selected designs, exact probabilities and source hashes accompany the aggregate snapshot.

E.3 Development selection and held-out prediction

We re-evaluate the originally selected VCC/Tahoe checkpoints at every recorded prefix (0, 1, 2, 4, 6, 8, 12, 16, 20 and 24), without refitting or reselection. The diagnostic reconstructs 14 distinct checkpoints and 140 development panels; saved held-out ranks independently reproduce all 80 task/backend/mode/prefix scores.

In VCC, each of the ten development panels contains two interventions; in Tahoe, each contains six. Their equal-panel macro Top-1 is consequently identical to the pooled intervention-macro Top-1: each development intervention receives weight $1/(10 \times 2)$ or $1/(10 \times 6)$. Different panel sizes in response-query counts do not change this identity, since the primary scorer first averages within an intervention. Thus a discrepancy between these two aggregation rules does not explain the reported decreases.

Development and held-out interventions are disjoint. Each intervention has a fixed five-option roster sampled within its split and shared by all methods. Selection therefore repeatedly uses a finite intervention set and its distractor contexts. Poor transfer to held-out interventions can reflect selection overfitting, biological shift or both; the original trajectories alone do not distinguish them.

VCC's 20 development interventions generate 302 response queries; its 20 held-out interventions generate 360. Tahoe has one response query per compound (60 development, 67 held-out). Qwen/VCC's

Table E.1: **Matched-access configuration search.** All methods use the same ten laboratories, development panels and 12-design library. Searches receive six proposal slots after the common initialization, with up to 700 aggregation rounds. Scores are multiplied by 100. Random search reports its exact expectation and search-policy standard deviation, integrating all 462 six-element subsets; this is not variation across training seeds.

Method	DTI AUROC↑	Proteomics AP↑	VCC Top-1↑	Norman Top-1↑	Tahoe Top-1↑
Fixed model	94.27	**37.27**	**31.50**	15.48	25.37
Random search, no replacement	94.45 ±0.17	**37.27** ±0.00	27.53 ±6.48	22.66 ±3.41	26.59 ±0.58
GP–EI Bayesian optimization	**94.60**	**37.27**	**31.50**	22.30	25.37
Qwen: direct search	94.27	**37.27**	**31.50**	22.30	25.37
Qwen: history-guided search	94.27	**37.27**	**31.50**	22.30	25.37
Luna: direct search	**94.60**	**37.27**	**31.50**	22.30	**26.87**
Luna: history-guided search	94.27	**37.27**	**31.50**	**23.11**	**26.87**

One frozen training seed per endpoint (42 for DTI; 61 otherwise); agent and GP–EI rows each represent one search trajectory. The fixed reference uses 100 rounds. Failed proposals retain their slots: Qwen history uses 600 fitted rounds on proteomics, and both Qwen arms use 500 on Tahoe; other searches use 700. Matched-realized-fit and shared-first-proposal sensitivities are reported separately. Selection uses development scores only; the comparison reuses checkpoint-identical, fully trained candidates and previously exposed held-out panels. Bold marks the largest displayed mean, not statistical significance.

Table E.2: Development–test divergence in the extended search (training seed 61). Each pair is initialization → 24 proposals. CPU re-evaluation reproduces the original development scores; test scores are independently recomputed from archived query ranks. The equal-laboratory development mean equals the pooled intervention-macro mean because each laboratory contains the same number of development interventions. All scores are macro Top-1 percentages.

Task	Proposer / search	Development	Held-out
VCC	Qwen / direct and history	23.65 → 24.83	31.50 → 19.61
VCC	Luna / direct	23.65 → 27.68	31.50 → 20.23
VCC	Luna / history	23.65 → 30.93	31.50 → 20.01
Tahoe	Both / direct and history	30.00 → 35.00	25.37 → 13.43

development gain of 1.17 points has a descriptive intervention-paired bootstrap interval of [−20.25, 21.54]; Tahoe's 5-point gain, equivalent to three additional correct development compounds, has interval [−11.67, 21.67]. These post-selection intervals describe the selected models and are not corrected for adaptive search. The primary-first rule can also accept worse loss: Qwen/VCC's development cross-entropy changes from 1.60783 to 1.61182, and Tahoe's from 1.60194 to 1.60877.

Development-roster sensitivity. Finite-sample selection criteria can themselves be overfit (Cawley & Talbot, 2010). We test one measurable contributor—dependence on the fixed option roster—without treating that general mechanism as a causal explanation of every observed decrease.

To test dependence on roster construction, we freeze 20 new roster seeds (2026110900–2026110919). Each true development intervention receives four distinct incorrect options sampled within its original development pool, and the five option positions are shuffled; all response queries for that intervention share the roster. Initial and already selected final checkpoints receive identical rosters. This study neither trains nor reselects a model and does not read test responses or scores.

Qwen/VCC's original advantage is positive in only 6 of the 20 new rosters; Tahoe's in 7. Luna/VCC retains smaller positive mean advantages, with 12 positive draws for each mode. The standard deviations quantify roster randomness on the same interventions, not independent training or biological replication. The result establishes sensitivity to distractors and option order; a new intervention-disjoint validation experiment would be required to separate it from biological distribution shift and repeated-selection effects.

E.4 What source-defined laboratories establish

Appendix B.2 establishes source transfer using PTPC, Lin, Ruprecht and decryptE for proteomics, and VCC, Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562 for cells. Proteomic AP improves

Table E.3: Development-roster sensitivity of the already selected initial and 24-proposal final checkpoints. Twenty frozen seeds resample four incorrect options within the same development-intervention pool; each comparison uses paired rosters. Entries are final-minus-initial macro Top-1 differences in percentage points. The standard deviation is across roster draws, not training seeds or new biological splits. No model is trained or reselected.

Task	Proposer / search	Original roster	New rosters	Positive draws
VCC	Qwen / both	$+1.17$	-1.34 ± 7.37	$6/20$
VCC	Luna / direct	$+4.03$	$+2.24 \pm 8.15$	$12/20$
VCC	Luna / history	$+7.28$	$+2.89 \pm 7.89$	$12/20$
Tahoe	Both / both	$+5.00$	-1.33 ± 4.82	$7/20$

Table E.4: Search with source-defined laboratories. Protein: PTPC, Lin, Ruprecht and decryptE; cell: VCC, Replogle, Nadig and Jiang. All methods share four training sources and the target development panels. Searches use twelve designs, six proposal slots and at most 700 rounds; the fixed reference uses 100 rounds. Mean $\pm$ sample SD over training/search seeds 61–63; scores $\times 100$. Random reports the exact policy expectation per seed, then its between-seed SD. Luna uses the same checkpoint cache but new proposal calls.

Method	Proteomic AP $\uparrow$	Cell macro Top-1 $\uparrow$
Fixed design	38.72 ± 1.94	28.82 ± 0.00
Random, exact expectation	36.37 ± 1.27	24.83 ± 4.23
Qwen direct	37.83 ± 1.28	22.36 ± 7.20
Qwen history	34.31 ± 4.96	19.41 ± 2.11
Luna direct	35.35 ± 3.62	22.88 ± 6.75
Luna history	34.94 ± 3.52	22.88 ± 6.75
Luna enriched history	35.39 ± 3.67	22.88 ± 6.75

from 34.41 to 37.53, whereas the matched cross-source cell setting scores 28.82 Top-1 versus 30.70 with same-source groups. These are simulations of different released studies, not independently deployed institutions. Table E.4 isolates proposal history at the same source access using complete twelve-design caches and three seeds. Its fixed reference belongs to this cache, rather than the separate seed-61 transfer study.

Qwen history scores below direct on both tasks. Adding expected-versus-observed feedback and configuration-change summaries to Luna history raises protein AP from 34.94 to 35.39, compared with direct's 35.35; all three cell modes select the same checkpoint within each seed. This enriched-context intervention changes several prompt fields and token costs. All 72 cached candidate predictions were reverified; the additional runs generate new proposals, not new predictor fits.

In these adapters, a target worker first averages ten target panels. The coordinator preserves that mean but sees one returning worker, so `worst_client` equals the primary mean rather than the worst panel. Selection remains unchanged, but history lacks cross-panel disagreement available in the main ten-client protocol.

E.5 Comparing training-promotion rules

The allocation result in Section 4.4 compares screening and promotion with uniform 80-round training. To examine the role of the promotion rule itself, we additionally compare the original score-plus-trajectory policy with three alternatives: promoting the six best screening scores globally, promoting the two best scores within each learning-rate group, and promoting six uniformly sampled designs. Screening comparisons use the best development primary score followed by lower development loss; complete screening ties retain the original slate order. Full-training selection retains the first promoted candidate on a complete score/loss tie. The trajectory policy retains the unchanged rule in Section 3.4.

All policies share ten designs, ten training laboratories, development panels and initialization rules. Promotion costs ten 20-round screens plus six fresh 100-round fits; uniform costs ten 80-round fits. Both use 800 aggregation rounds and 160 development evaluations, with design-dependent runtimes. We train every design afresh at each duration for four tasks and seeds 53–55: 360 fits and 24,000

actual rounds. This complete cache supports retrospective policy evaluation; its construction cost is separate from each policy's budget.

Table E.5: Fresh-runtime allocation controls. Scores $\times 100$ are mean $\pm$ sample SD over training seeds 53–55. All policies use the same ten laboratories and ten designs, with 800 aggregation rounds and 160 aggregate development evaluations. Promotion policies screen ten designs for 20 rounds and restart six for 100; uniform allocation fits all ten for 80. Random promotion integrates all 210 six-design subsets per seed, and its displayed SD is across seed-conditional expectations. All 360 fits are newly trained; the original Table 2 is retained separately. Bold marks the best unrounded mean, including ties.

Allocation policy	Proteomics AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
Uniform (80 rounds)	**37.47** $\pm$ 0.55	20.01 $\pm$ 0.00	11.48 $\pm$ 0.52	20.90 $\pm$ 2.99
Random promotion	37.37 $\pm$ 0.48	**22.49** $\pm$ 1.70	**22.63** $\pm$ 0.64	21.95 $\pm$ 0.04
Score only (global)	**37.47** $\pm$ 0.55	20.01 $\pm$ 0.00	21.70 $\pm$ 7.19	20.90 $\pm$ 0.00
Score only (LR groups)	**37.47** $\pm$ 0.55	20.01 $\pm$ 0.00	21.70 $\pm$ 7.19	20.90 $\pm$ 0.00
Score + trajectory	**37.47** $\pm$ 0.55	20.01 $\pm$ 0.00	22.32 $\pm$ 0.43	**22.89** $\pm$ 1.72

Development selections are sealed before held-out scoring. Random promotion integrates all $\binom{10}{6} = 210$ subsets, with original slate order breaking final ties. Its reported SD is across three seed-conditional expected single-predictor scores; within-seed policy variation is recorded separately. This fresh-runtime comparison uses the original biological partitions and preserves the original study in Table 2.

The fresh run reproduces all original uniform and trajectory means. Across twelve task–seed pairs, trajectory promotion yields 4/8/0 wins/ties/losses against uniform, 4/7/1 against either score-only rule and 6/0/6 against random expectation. These are paired directions, not pooled effects across metrics. Both score-only rules give the same task means. Norman's trajectory score exceeds theirs by 0.62 points but trails random; Tahoe favors trajectory, while VCC favors random. Fully training fewer candidates explains much of the gain over uniform allocation, with task-dependent effects from the promotion rule.

E.6 History representation on a larger gene-disjoint study

The preceding controls motivate a fresh experiment that separates feedback representation from access and fitting budget. It uses two source studies, Replogle and Nadig, with four study–cell contexts: K562, RPE1, HepG2 and Jurkat. Each context acts as a training and evaluation laboratory. These are four biological contexts, not four independently deployed institutions. After vocabulary and measurement filtering, the training, development and held-out sets contain 3,806, 1,247 and 1,282 conditions, respectively, covering 1,208, 414 and 406 distinct perturbation genes. Gene names and model-vocabulary identifiers are disjoint across the three partitions.

Predictor and measurement. We freeze the corrected scDEBART encoder and select 512 response genes by training-response variance, requiring measurement in all four contexts. The predictor receives control expression and the candidate perturbation identifier. It predicts the difference between perturbed and control mean log-normalized expression. Evaluation identifies the true perturbation among five options within a context. Each gene–context is evaluated on five fixed rosters; every option uses the same mask, excluding the five candidate perturbation genes wherever they occur in the response panel. The primary score first averages rosters within a gene, then genes within a context, and finally the four contexts equally. Repeated rosters do not create independent biological replicates. This new response panel and source-defined benchmark are separate from the original VCC, Norman and Tahoe endpoints.

Common candidate and fitting protocol. The registered library has 16 designs: four learning rates ($10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}$), two prediction heads (fine-tuned pretrained head or frozen head plus a zero-output-initialized residual adapter), and two objectives (response MSE or MSE plus 0.1 times within-batch contrastive cross-entropy). Weight decay is 0.001 and coordinator momentum is zero. All 48 design–seed fits start afresh and complete 20 full-client rounds with batch size 32. Development checkpoints at rounds 2, 5, 10 and 20 are ranked by Top-1, then lower cosine-ranking

cross-entropy and earlier round. The pretrained round-0 predictor is diagnostic only. The fixed D00 reference is the fully trained head with learning rate 10^{-4} and MSE, not the untrained initialization.

Matched feedback conditions. Qwen2.5-7B-Instruct generates six candidate requests after D00, with the first request shared across three modes. Direct receives tried identifiers without scores; raw history receives the observed development scores, trajectories and context diagnostics; structured history receives the same facts organized around reference configurations, exact configuration changes and expected-versus-observed effects. Neither history mode observes an unqueried candidate. Proposals contain `design_id`, `reference_id`, `changed_fields`, the hypothesis, experiment and expected effect, and a numeric prediction. The executor checks identifiers and configuration changes against the registered library; those fields, rather than the rationale text, determine execution. This makes the proposal's stated comparison explicit without assigning credit merely for predicting an outcome correctly.

The final study reruns all three modes after an earlier output parser rejected otherwise valid JSON wrapped in code fences or containing literal backslashes. The same limited formatting normalization is applied to every mode; malformed schemas and incorrect configuration changes still fail, with at most two generation attempts per slot. The earlier run remains an engineering diagnostic, not an additional scientific replicate. Direct and structured history each complete all 18 slots across seeds 61–63; raw history completes 15. Conventional random search and GP–EI use the same candidate cache and development-selection key. Random search exactly integrates the $\binom{15}{6} = 5,005$ subsets, with shared-first and realized-fit-count sensitivities retained in the result record.

Table E.6: Fresh history-feedback comparison on the gene-disjoint, four-context cell benchmark (three paired training/controller seeds). This benchmark is separate from the original VCC, Norman and Tahoe tasks. Top-1 scores are percentages; $\pm$ reports the sample standard deviation across seeds. Search methods share the same 16 designs, four clients, development panels and six proposal slots; each valid candidate is fitted for 20 rounds. Fits include the initial-design fit. Random search integrates all size-six subsets of the 15 non-initial designs and selects using development evidence, not test scores. Raw history has three invalid slots across 18; direct and structured history complete all 18. Bold marks the best mean development or test score.

Search method	Development	Held-out	Fits
Fully trained fixed D00	22.40 ± 0.07	21.43 ± 0.10	1
Direct (no feedback)	22.68 ± 0.25	22.62 ± 0.21	7
Raw history	22.68 ± 0.25	22.62 ± 0.21	6 ± 1
Structured history	23.38 ± 0.48	22.89 ± 0.56	7
Random without replacement	23.06 ± 0.24	22.67 ± 0.21	7
GP–EI	$\mathbf{23.60 \pm 0.32}$	$\mathbf{23.15 \pm 0.35}$	7
Paired held-out contrasts (percentage points)			
Contrast	Mean difference	95% CI	W/T/L
Structured − Direct	$+0.263$	$[-1.022, 1.568]$	2/1/0
Structured − Random	$+0.220$	$[-0.290, 0.738]$	2/0/1
Structured − GP–EI	-0.264	$[-0.930, 0.413]$	0/2/1

W/T/L counts paired seeds. Confidence intervals use 10,000 paired perturbation-gene bootstrap resamples and describe conditional test-gene uncertainty for the fixed selected models, rather than uncertainty over additional training or controller runs.

Outcomes and resource accounting. Structured history selects a different final design from direct in two of three seeds. Its mean held-out Top-1 is 22.888%, versus 22.625% for direct and 22.668% for random search; GP–EI reaches 23.152%. Structured-minus-direct is $+0.263$ points (two gains and one tie), with paired gene-cluster 95% interval $[-1.022, 1.568]$. Raw history and direct select the same final predictor in every seed. The confidence intervals use 10,000 global-gene resamples shared across contexts, methods and seeds, conditional on these fitted and selected predictors. They measure held-out gene uncertainty, not variation over further controller runs. The results show that the feedback representation can change search decisions, but do not establish stable superiority over direct or conventional search.

Every valid fit uses 2,420 optimizer steps and 76,120 training-example exposures. Direct, structured, random and GP–EI therefore each use 16,940 steps and 532,840 exposures per search. These matched

Table E.7: Resource accounting for the fresh history comparison. Training columns are per-seed means; the last two columns total all three seeds. Fits are evaluated from a shared cache, with each policy charged for every candidate it requests. Raw-history invalid slots consume proposal budget without a candidate fit. Identical rounds, steps and examples do not imply identical FLOPs or language-model costs. Calls include permitted retries and logically charge the shared first proposal to each language-model arm.

Method	Rounds	Steps	Examples	Invalid slots	LLM calls
Fully trained fixed D00	20	2,420	76,120	0	0
Direct (no feedback)	140	16,940	532,840	0	18
Raw history	120	14,520	456,720	3	24
Structured history	140	16,940	532,840	0	21
Random without replacement	140	16,940	532,840	0	0
GP–EI	140	16,940	532,840	0	0

counts do not imply identical FLOPs: the head choices have different trainable parameter counts. Logical language-model calls total 18 for direct, 24 for raw and 21 for structured, including retries; generating the shared first proposal only once gives 57 physical calls overall. Training/development feature extraction and all 48 predictor fits cost approximately 0.236 GPU hours, excluding language-model inference and held-out evaluation. This construction cost is distinct from each policy's logical budget. Held-out scoring follows the frozen development selections and independently verifies all 48 models.

Development and held-out trajectories. Figure 6 reports every prefix from zero to six requests for every policy and seed. All final search outcomes improve on their corresponding fitted D00 reference, so this study does not reproduce the earlier large terminal decreases. It does retain a local reversal: in seed 62, raw and structured history select D05 at slot 2, raising development Top-1 by 0.057 points while lowering held-out Top-1 by 0.257; later selections recover. The K562 context also has a positive mean development change but a negative mean held-out change under structured history. Its largest overall development improvement occurs at the final slot, and its prefix-average development score remains below random search and GP–EI. The new data, response panel, design library and feedback format change together relative to the old benchmark, so the improved terminal behavior is not a controlled attribution to any one change. The diagnostic scores the originally selected prefix models; it does not choose a stopping point using held-out outcomes.

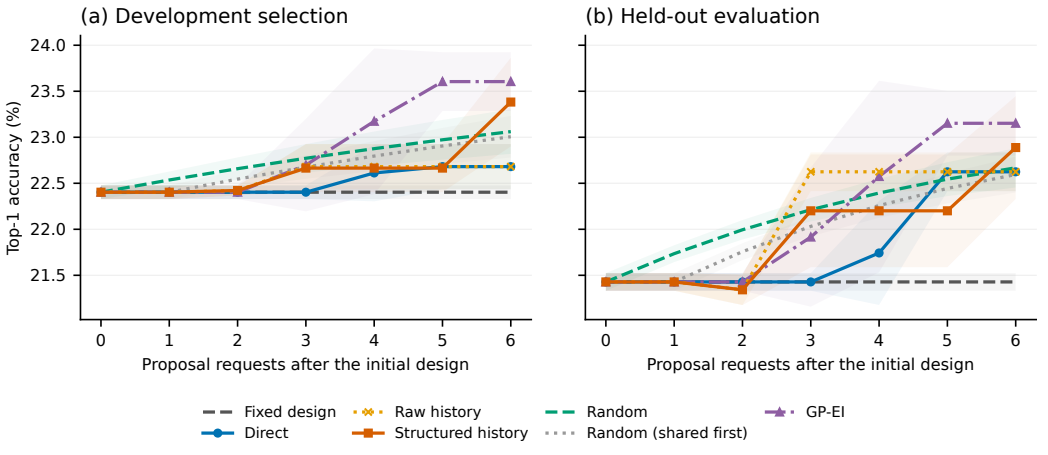


Figure 6: Development selection and held-out prediction across the complete six-request study. Lines are means across three paired seeds and shading is sample standard deviation. Random curves integrate development-selected single-model outcomes at each budget, rather than depicting one random trajectory. The fixed reference has received its full 20-round fit. Held-out trajectories evaluate the development-selected model at each prefix, not the best model according to test scores.

Table E.8: Fixed-design sensitivity to the identity of the single participating laboratory. Each of ten laboratories independently fits D00 for 100 rounds and selects a checkpoint on its own development panel (seed 42). All 40 fits are fresh CPU runs under the same training equations; the SD and range describe laboratory identities, not training seeds. This diagnostic does not rerun research controllers.

Endpoint	Lab 0	Single-lab mean $\pm$ SD	Range
Proteomics AP	24.01	28.69 ± 6.44	14.69–37.34
VCC Top-1	29.42	32.50 ± 3.49	26.96–39.76
Norman Top-1	25.56	17.67 ± 5.84	4.37–25.56
Tahoe Top-1	19.40	19.70 ± 4.66	13.43–26.87

E.7 Sensitivity to the single participating laboratory

To examine whether laboratory 0 represents independent laboratories generally, we additionally train the fixed D00 design separately on every laboratory, 0–9, for proteomics, VCC, Norman and Tahoe. Each fit uses seed 42, 100 rounds, only that laboratory's training data and only its development panel for checkpoint selection. All 40 models, including laboratory 0, are newly fitted under the same CPU runtime and sealed before evaluation on the common task-specific test set. Saved predictions independently reproduce all 40 primary scores. The standard deviation here describes variation across laboratory identities at one seed, not training-seed variation or a collaborative model.

Laboratory 0 is below the independent-laboratory mean for proteomics and VCC, close to it for Tahoe, and the highest-scoring laboratory for Norman. Thus its representativeness is task-dependent. These results supplement, rather than replace, the original three-seed GPU workflow scores. They evaluate fixed-model identity sensitivity, not a rotation of every search controller across laboratories. DTI is not included: its present preprocessing uses a training dictionary constructed from laboratory 0, so an isolated-laboratory rotation would also require a laboratory-specific dictionary and a separately registered initialization protocol. The same-ten-laboratory search controls in E.1–E.2 remain the direct comparison of search methods at fixed access.